Who Holds U.S. Bonds and How It Shapes the Yield Effects of QE and QT

Masashi Takahashi*†

November 10, 2026

[Click here for the latest version]

Abstract

How much do the bond-supply shifts of quantitative easing (QE) and tightening (QT) move Treasury yields and mortgage rates? The answer turns on the slope of aggregate bond demand: who holds bonds, how sensitive their demand is to yields, and how fast they rebalance. I estimate heterogeneous investor demand for Treasuries and the agency MBS that fund most U.S. mortgages, using sector-level portfolio holdings and transactions from 1985 to 2025, measuring each sector's demand against its own non-bond alternative and letting its yield sensitivity and rebalancing speed vary. Most investors respond to yields weakly and slowly, so aggregate demand is inelastic: historical QE compressed 10-year Treasury yields by 8 to 12 basis points per percent of the float removed. Further runoff would move yields less per unit, because the sectors that absorb Treasuries now hold more of the float: an additional $1 trillion raises 10-year yields by about 22 basis points and agency MBS yields by about 27. Agency MBS demand is far more elastic than Treasury demand, so the mortgage cost of QT comes through the Treasury yield, not the mortgage spread.

Keywords: Agency MBS, Demand System, Institutional Investors, Mortgage Rates, Quantitative Easing, Quantitative Tightening, U.S. Treasuries

*Smeal College of Business, Pennsylvania State University. Email: mft5548@psu.edu.

†I am grateful to Brent Ambrose and Jiro Yoshida for their guidance and support. I also thank Sriya Anbil, Daniel Beltran, Cynthia Doniger, Erin Ferris, David Glancy, Hyein Han, Jingzhi Huang, Kyungmin Kim, Robert Kurtzman, Anh Le, Stefan Lewellen, Canlin Li, Liang Peng, Lubomir Petrasek, Romina Ruprecht, Bernd Schlusche, Tim Simin, Arsenios Skaperdas, Eva Steiner, Hiroatsu Tanaka, Manjola Tase, Mihail Velikov, Annette Vissing-Jørgensen, Sean Wilkoff, Alexei Zhdanov, and seminar participants at the Federal Reserve Board and Penn State for their helpful comments and discussions. Needless to say, any remaining errors are solely my responsibility.

1 Introduction

How much does the Federal Reserve's balance-sheet policy move Treasury yields and mortgage rates? Over more than a decade, the Fed purchased more than $6 trillion of U.S. Treasuries and agency MBS to lower yields, a policy known as quantitative easing (QE). It has since reversed course, shrinking its balance sheet through quantitative tightening (QT), yet the consequences for Treasury yields and mortgage rates, and the channel through which they operate, remain unclear.[1] The key is that QE and QT shift the supply of bonds available to private investors, so their effect on yields depends on the slope of investor demand curves: flat demand implies small yield effects, steep demand large ones (e.g., Koijen and Yogo, 2019; Vayanos and Vila, 2021). Because investors differ in their yield sensitivity, in the assets they hold, and in how quickly they rebalance, the aggregate demand curve, and hence the size and timing of the yield effect of QT, depends on *which* investors absorb the released supply, and how fast.

This paper estimates bond demand using sector-level portfolio holdings and transactions covering the two bond markets the Federal Reserve trades, Treasury securities and agency MBS, across 27 sectors grouped into eight broad sector types: banks, insurers, pension funds, mutual funds, arbitrageurs, foreign investors, households, and other investors. Four choices define the system. I measure each sector's demand from its transactions rather than from the market value of its holdings. I measure it against the non-bond asset each sector actually trades off against bonds, its own outside asset. I let each sector's yield sensitivity differ by asset class, so the system can say who absorbs Treasuries and who absorbs agency MBS instead of imposing one slope per investor across markets. And I let each sector rebalance at its own speed, estimating how far and how fast each adjusts. The estimates support counterfactual analyses of the QE and QT episodes and of the runoff to come.

I find three main results. First, most investors are inelastic, and the few that absorb

[1] In June 2026, Federal Reserve Chairman Kevin Warsh made balance-sheet policy one of five priorities for the conduct of monetary policy, charging a dedicated task force to weigh "the benefits and risks of the current ample reserves regime, and the composition of the Fed's balance sheet" (Warsh, 2026).

supply do so slowly. In the Treasury market only two sectors, holding under a third of the float, have slopes distinguishable from zero: the household residual, after 2010 mostly hedge funds in the cash–futures basis trade, and mutual funds. In the agency MBS market, by contrast, every sector type's demand rises with the spread over Treasuries, and for all but mutual funds more strongly than with the Treasury yield: agency MBS is a spread product held as an investment, whereas most Treasuries are held for liquidity, reserves, and mandates by investors who do not trade on the yield. Second, with so few absorbers, historical QE compressed 10-year Treasury yields by 8 to 12 basis points per percent of the float removed, in line with the demand-system literature (Eren et al., 2026; Chaudhary et al., 2026). Because the Treasury market sets the level of agency MBS yields and the MBS market only the spread, Treasury purchases lower agency MBS yields too: QE2, a Treasury program, lowered them by 90 basis points, while QE1, the only MBS-heavy program, compressed the spread by a further 54. Third, a resumed QT raises mortgage costs mostly through the Treasury market. A further $1 trillion of runoff from the 2025Q1 balance sheet, about a year and a half at the Fed's 2024 pace and split roughly two-thirds Treasuries and one-third agency MBS as in the current runoff, widens the mortgage spread by only 5 basis points but raises 10-year Treasury yields by 22, so agency MBS yields rise by 27, with a wide band on the Treasury number (Section 5).

I construct quarterly sector-level portfolio shares and transactions for the two bond classes from the Federal Reserve's Financial Accounts (Z.1), splitting the rest of the world into official and private holders with the Federal Reserve's CSLT cross-border positions data over the full sample. The Financial Accounts report every holding together with the net purchases that changed it, decomposing each level change into transactions, revaluations, and other volume changes. Transactions measure what a sector chose to buy or sell; the change in its holdings also carries the revaluation of what it already held, which moves with the yield itself, so measured from market values an investor that traded nothing would appear to sell when yields rise. A duration-based correction (Eren et al., 2026) presumes

every holder records that revaluation, which banks and insurers at book value do not; using the revaluation each sector itself records avoids the problem. I merge the transaction panel with asset characteristics and macroeconomic controls from FRED. The yield is the 10-year Treasury yield for Treasuries and, for agency MBS, the 10-year yield plus the index option-adjusted spread (OAS), whose spread component is the price MBS demand responds to; maturity, duration, and coupon come from the Bloomberg indices, and issuance from the Financial Accounts. The sample covers 27 sectors and two asset classes over the 161 quarters from 1985Q1 to 2025Q1.

Given these data, I estimate a characteristics-based demand system following Koijen and Yogo (2019), in which each sector's portfolio shares take a logit form, and extend it in four ways. First, the yield coefficient differs by asset, the alternative-specific-coefficient logit of discrete choice; the share structure and its closed-form elasticities are unchanged, and a minimum-distance test rejects the equal-slope restriction (Appendix D.8). Second, I measure demand from transactions rather than from market values, which keeps the revaluation of positions never traded out of the slopes. Third, because sectors rebalance at different speeds, I estimate the demand equation in first differences of transaction growth and trace the response horizon by horizon with local projections (Jordà, 2005), recovering each sector's impact slope, long-run slope, and speed of adjustment. Fourth, the outside asset is sector-specific, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers, so the slope captures substitution between bonds and each holder's realistic alternative. The slope of each market's aggregate demand curve, the float-weighted sum of the holders' own-asset slopes, aggregates across sectors whatever outside asset each uses, and it is the object the counterfactual requires.

Yields are endogenous, and I identify each asset's slope with instruments specific to that asset. For both markets the Fed trades I use the revisions to its *announced* purchase path rather than its realized purchases, since the realized flow follows a schedule published in advance and carries no surprise. The instrument moves the mortgage spread while leaving the

Treasury yield untouched, the test an asset-specific supply shifter has to pass. For Treasury yields I add a heteroskedasticity-based instrument (Lewbel, 2012) built on the coupon. I set aside the quarters of the flight-to-quality episodes documented in the literature, from the 1987 crash to the 2022 tightening, when flight-to-quality demand shocks move yields directly: the regressions control for them, and the heteroskedasticity instrument is built only from calm-period variation. The first-stage F statistics are 7.7 for Treasuries, on two instruments whose overidentifying restriction is not rejected, and 18.1 for agency MBS, and I report a weak-instrument-robust Anderson–Rubin set next to every slope, describing a slope as different from zero only when its set excludes zero.

The estimated demand parameters show who absorbs supply in each market. In the Treasury market only two sector types have long-run slopes whose Anderson–Rubin sets exclude zero, the household residual and mutual funds; banks, insurers, pension funds, and foreign investors, who hold most of the float, have slopes whose sets include zero. In the agency MBS market four of the eight do: the household residual, foreign investors, banks, and pension funds. Absorption is slow in both markets. A typical absorber completes only an eighth to a quarter of its one-year response in the quarter of the shock, pension funds being the one fast absorber, and several absorbers are still building a year later, the slow-moving capital of Duffie (2010) as a property of every sector that absorbs supply. In agency MBS the household residual and the arbitrageurs even trade the other way on impact, so that market has no marginal buyer in the quarter a supply shock lands.

The demand slopes and the composition of holdings together determine the slope of each market's aggregate demand curve, which pins down the basis-point impact of a supply shift. In 2025Q1 the Treasury slope implies that removing 1 percent of the private float raises the 10-year yield by about 8 basis points, with a 95 percent band of 4 to 29, and the household residual supplies more than four-fifths of the slope behind it. The agency MBS slope implies that removing 1 percent of the MBS float widens the mortgage spread by 2.0 basis points, with a band of 1.3 to 4.2, and unlike the Treasury slope it rests on many sector types rather

than one. The Treasury slope has moved with that one sector's presence: it fell by two-thirds in the early 2000s and stayed low through 2008, as foreign reserve managers accumulated Treasuries and the residual emptied, and it recovered after 2009 and again after 2022 as the float shifted back toward the residual and the hedge funds in it.

Lastly, I conduct a counterfactual analysis of Fed asset purchases using the estimated system. Equilibrium yields are jointly determined by the demand system and the market-clearing condition in the two markets the Fed trades, with bond supply responding to yields through net-issuance elasticities estimated from the same data. Across the four programs, QE1 through QE4, the Treasury impact runs from 8 to 12 basis points per 1 percent of the float removed, close to the 8 to 13 of Eren et al. (2026). QE1's mortgage effect of 108 basis points sits just above the event-study range of 85 to 107. The 2017–2019 runoff raised Treasury yields by 23 basis points and agency MBS yields by 31. The 2022–2025 runoff raised them by 65 and 76. Looking forward, a further $1 trillion raises 10-year Treasury yields by 22 basis points and agency MBS yields by 27; applied to $27 trillion of marketable federal debt as it rolls over, the Treasury rise adds roughly $60 billion a year to federal interest costs, and the mortgage rise about $72 a month to the payment on a newly originated $400,000 mortgage. The Treasury number is a band, not a point: it runs from 6 basis points if the runoff lands in a recession to 45 at the lower edge of the household slope's confidence set.

This paper contributes to the literature on market macrostructure (Haddad and Muir, 2025), and in particular to demand-system asset pricing in bond markets.[2] Koijen et al. (2021) find that foreign investors, the most elastic holders, sell the most to the ECB during quantitative easing. Jansen et al. (2024) find that QE moves Treasury yields only insofar as investors perceive the purchases as persistent, the case my counterfactuals correspond to, and that hedge funds and dealers are the marginal Treasury investors, the high-frequency counterpart of the residual sector that absorbs Treasuries here. Eren et al. (2026) find asymmetric yield effects of QE and QT for U.S. Treasuries. Fang and Xiao (2025) estimate

[2]Several papers estimate demand systems for bond markets without a direct focus on QE: Fang et al. (2025) for global sovereign debt and Bretscher et al. (2025) for corporate bonds.

three bond classes at the security level, find that mutual funds and banks buy a further 0.5 units for every unit the Fed buys, and recover substitution between long Treasuries and agency MBS that my sector-level classes average over. Chaudhary et al. (2026) find aggregate Treasury demand highly inelastic, 13 basis points per 1 percent shift in demand, close to the upper end of the per-unit impacts estimated here. My contribution is a detailed evaluation of QE and QT through a demand system that covers every holder of the two markets, 27 sectors across which the float clears: I price every program from QE1 to QT2 and a prospective runoff in both markets the Fed trades, separate the Treasury level from the mortgage spread so that a program's composition can be read off its effect on mortgage rates, and trace how the yield effect builds and settles as slow capital arrives. The four features described above, transactions, sector-specific outside assets, asset-specific slopes, and sector-specific speed, are what make this possible.

Event studies find that the Fed's purchase announcements lowered 10-year Treasury yields by 91 to 107 basis points around QE1 and primary mortgage rates by 40 to 100 (Fuster and Willen, 2010; Gagnon et al., 2011; Hancock and Passmore, 2011; Krishnamurthy and Vissing-Jorgensen, 2011; Stroebel and Taylor, 2012; Krishnamurthy et al., 2013), though announcement effects mix signaling with portfolio balance (Christensen and Rudebusch, 2012); term-structure studies map the Fed's holdings into term premiums at roughly 30 to 45 basis points per program (D'Amico et al., 2012; Ihrig et al., 2018; Wei, 2022), and actual purchases show persistent stock effects (D'Amico and King, 2013; Vissing-Jorgensen, 2021). These estimates evaluate only the policies implemented, under the conditions of the time, and cannot price alternative sizes or asset mixes. Demand parameters defined independently of the policy can, subject to the stability caveats of Sections 4 and 5. Calibrated preferred-habitat models also support counterfactuals (Vayanos and Vila, 2021); I estimate demand from observed sector transactions and cover both markets the Fed trades, so the counterfactuals inherit measured investor heterogeneity.

QE transmits through signaling, portfolio rebalancing, and asset-specific channels (Kr-

ishnamurthy and Vissing-Jorgensen, 2011), bank lending (Chakraborty et al., 2020; Kandrac and Schlusche, 2021), and mortgage refinancing (Di Maggio et al., 2020). This paper quantifies portfolio rebalancing, each sector's substitution between the bonds the Fed trades and its outside asset, and speaks to the debate between a broad duration channel, under which purchases in any market move all long-term yields (Gagnon et al., 2011), and the narrow view that the impact concentrates in the purchased asset (Krishnamurthy et al., 2013). Each holds in part: Treasury purchases lower agency MBS yields through the Treasury level, while the mortgage spread responds mainly to the MBS the Fed buys or runs off.

Finally, how far the central bank can shrink its balance sheet depends on both sides of it. On the liability side, QT drains reserves and strains the repo market, so the demand for reserves bounds its size (e.g., López-Salido and Vissing-Jorgensen, 2025; Anbil et al., 2026; Anderson et al., 2026; Oguri and Pizzimenti, 2026). This paper studies the asset side: investors take the bonds the Fed releases only if yields rise, and that rise in yields is a second limit on QT. That effect cannot be read off QE. The asymmetry arises on the liability side from reserve scarcity (Acharya et al., 2024) and on the asset side from the composition of the float, which shifts who must absorb the supply, and from the state of the cycle in which the runoff lands. Jiang and Sun (2024) make the asset-side point in a calibrated model with slow-moving long-term investors, in which a gradual runoff limits the overshoot; I estimate the slopes and adjustment speeds such a model calibrates. Assessing QT requires knowing who holds bonds, which of them trade on yields, how steeply, and how fast; for Treasuries the answer today rests on two sectors, one of which supplies four-fifths of the slope.

2 Data

The dataset covers the information required for the demand estimation: portfolio holdings and the transactions that change them, asset prices (yields), asset characteristics, and instrumental variables for asset prices.

2.1 Portfolio Holdings and Transactions

Sectors. I use the Federal Reserve's Flow of Funds (the Financial Accounts of the United States, Z.1) to compute sector-level portfolio shares and to measure the transactions behind their changes. The Flow of Funds reports quarterly balance sheets for the U.S. financial sectors (e.g., life insurance companies, private pension funds). The sample runs from 1985Q1 to 2025Q1, a span that reaches back to the high-yield years of the late 1980s and, as Section 3.7 shows, is what identifies the marginal buyer of Treasuries. Over this span I work with 27 sectors: the 25 domestic sectors of the Financial Accounts and the rest of the world split into Foreign Official and Foreign Private holders. The split uses the Federal Reserve's CSLT estimates of cross-border securities positions (Bertaut and Judson, 2014), which report monthly holdings and net transactions of U.S. Treasuries, agency bonds, corporate bonds, and equities by official and non-official foreign investors from 1984 onward. A separate hedge-fund sector from the Federal Reserve's Enhanced Financial Accounts (EFA), built from SEC Form PF filings, is available only from 2013, so the sample keeps hedge funds within the household residual; Appendix D.6 uses the EFA to measure their share of it. Table 1 lists the sectors and their broader sector-type classifications (e.g., insurers, pension funds); the foreign official and private units both belong to the foreign type.

Assets. The two bond markets the Federal Reserve trades, Treasury securities and agency MBS, are the assets inside the demand system; the Flow-of-Funds asset classes are coarser than CUSIP-level data but detailed enough (e.g., Treasury securities, corporate equity, cash) to identify them. Together the sectors form a heterogeneous demand side, yet a few large groups hold most of the bonds: on average over the sample, foreign investors and domestic financial institutions (banks, insurers, pension funds, and mutual funds) hold about 80 percent of the two markets, while arbitrageurs (dealers and mortgage REITs) hold only about 5 percent; hedge funds sit inside the household residual. Figure 1 reports each market's ownership composition over time, from 1985 to 2025. Because the 10-year yield prices

Table 1: **Sector Coverage and Outside Asset Definitions**

Type	Sector	Outside Asset
Banks	U.S.-Chartered Depositories	Loans
	Foreign Banking Offices	Total non-bond
	Credit Unions	Loans
	Banks in Affiliated Areas	Total non-bond
Insurers	Life Insurers	Corporate equities
	Property-Casualty Insurers	Corporate equities
Pension Funds	Private Pension Funds	Corporate equities
	State & Local Govt. Retirement Funds	Corporate equities
	Federal Govt. Retirement Funds	Total non-bond
Mutual Funds	Mutual Funds	Corporate equities
	Exchange-Traded Funds	Corporate equities
	Money Market Funds	Repo
	Closed-End Funds	Corporate equities
Arbitrageurs	Brokers and Dealers	Repo
	mREITs	Total non-bond
Foreign	Foreign Official	U.S. equities
	Foreign Private	U.S. equities
Household (incl. hedge funds)	Households & Nonprofits	Corporate equities
Other	Nonfinancial Business	Total non-bond
	Government-Sponsored Enterprises	Total non-bond
	Holding Companies	Total non-bond
	State & Local Governments	Total non-bond
	Federal Government	Total non-bond
	Agency/GSE Mortgage Pools	Total non-bond
	Finance Companies	Total non-bond
	ABS Issuers	Total non-bond
	Other Financial Business	Total non-bond

Notes: This table shows the sectors in the portfolio data and the outside asset against which each sector's bond demand is measured, the non-bond asset it trades off against bonds; "Total non-bond" is total financial assets net of all bond holdings. The rest of the world is split into official and private holders with the Federal Reserve's CSLT data; hedge funds stay inside the household residual (Appendix D.6). Appendix D.1 explains each outside asset and Appendix Table A.1 lists the sources.

coupon securities rather than bills, I measure the Treasury asset on the coupon market, using the non-bill holdings and transactions the Accounts publish for the six holders whose bills they break out (tables L.210 and F.210) and the total for the remaining sectors. Agency MBS is left as reported: table L.211 does not separate agency MBS from the debentures of the government-sponsored enterprises, about 17 percent of the total, which attenuates the estimated MBS slopes.

Transactions and Revaluation. The change in a sector's portfolio mixes two things: the sector's trading, and the revaluation of the positions it already holds when prices move. Demand estimation has to separate the second from the first. Eren et al. (2026) approximate the revaluation with a duration-based price change of the bond index. The Financial Accounts also record each sector's revaluation of every instrument alongside its net purchases, so I can use the revaluation each sector reports directly. A recorded holding evolves as

$$H_{it}(n) = H_{i,t-1}(n) + F_{it}(n) + R_{it}(n), \tag{1}$$

where $H_{it}(n)$ is sector i's recorded holding of asset n, $F_{it}(n)$ its net purchases during quarter t, and $R_{it}(n)$ the revaluations and other volume changes. Taking logs, the growth of the holding splits exactly into a trading term and a revaluation term,

$$\ln \frac{H_{it}(n)}{H_{i,t-1}(n)} = \underbrace{\ln\Big(1 + \frac{F_{it}(n)}{H_{i,t-1}(n)}\Big)}_{g_{it}(n),\ \text{trading}} + \underbrace{\ln\Big(1 + \frac{R_{it}(n)}{H_{i,t-1}(n) + F_{it}(n)}\Big)}_{r_{it}(n),\ \text{revaluation}}, \tag{2}$$

and so does the change in the log share ratio against the outside asset:

$$\Delta \ln \frac{H_{it}(n)}{H_{it}(0)} = \big[g_{it}(n) - g_{it}(0)\big] + \big[r_{it}(n) - r_{it}(0)\big]. \tag{3}$$

I remove the recorded revaluation and use the remainder as the dependent variable of the demand system:

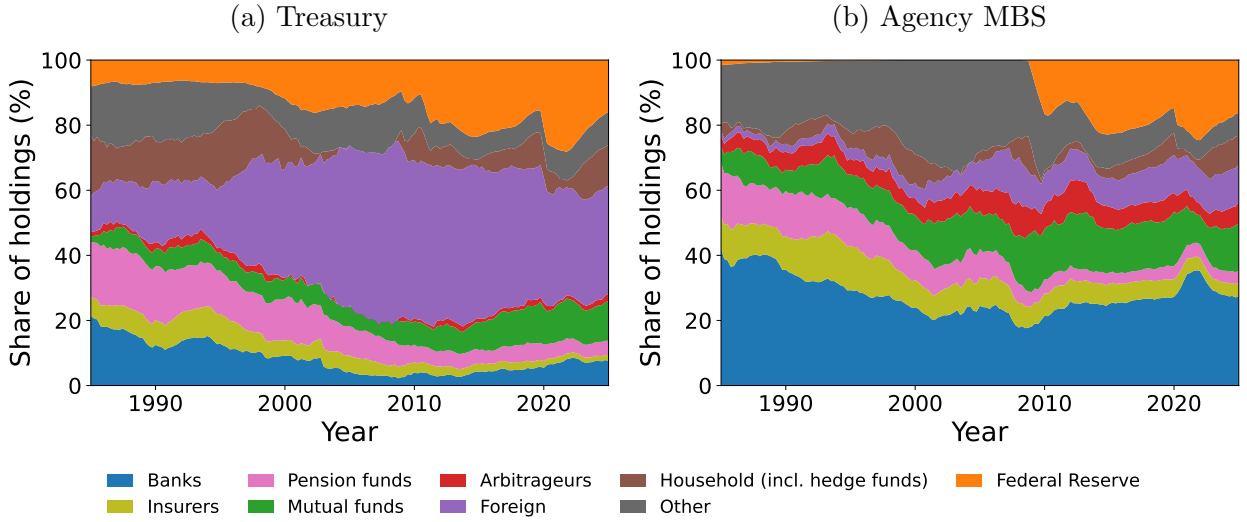
$$\Delta q_{it}(n) \equiv \Delta \ln \frac{H_{it}(n)}{H_{it}(0)} - \big[r_{it}(n) - r_{it}(0)\big] = g_{it}(n) - g_{it}(0). \tag{4}$$

The two approaches coincide for a holder that marks its book to market, and the recorded flows add sector detail where accounting conventions differ: banks report held-to-maturity securities at amortized cost, insurers at statutory book value, and the rest of the world at market value, so the recorded pass-through of a price change into the level ranges from zero to nearly one across sector–asset cells (Appendix D.2). With the recorded revaluation removed, a sector that neither buys nor sells has $\Delta q_{it}(n) = 0$ whatever its accounting. I winsorize each cell's transaction growth at its 1st and 99th percentiles, because a growth rate is uninformative on a near-zero base, and leave it undefined where the lagged position is below $50 million.

Outside Asset. The outside asset is sector-specific: for each sector I designate the single non-bond class it trades off against bonds, and measure its bond demand relative to that class. Domestic banks, for example, rebalance between bonds and loans rather than between bonds and every other asset on their balance sheet, so loans are their outside asset. The last column of Table 1 reports each sector's outside asset, and Appendix D.1 gives the rationale sector by sector. Appendix Figure A.1 shows how each sector's portfolio shares evolve over the sample.

Limitations. I acknowledge the limitations of the sector-level portfolio share data from the Flow of Funds. The data do not report within-class maturity variation, so I cannot estimate demand across precise duration buckets. Thus, this paper focuses on substitution across broad fixed-income asset classes rather than fine maturity choice.

Figure 1: **Ownership Structure of U.S. Bond Markets**



Notes: This figure shows the ownership shares of the coupon Treasury market and the agency MBS market by sector type (Table 1), including the Federal Reserve, from the Z.1 Financial Accounts. The household residual's Treasury holdings are mostly hedge funds in the cash–futures basis trade after 2010 (Appendix D.6); the residual is floored at zero where the Accounts report it as negative (2004Q2–2008Q2).

2.2 Asset Characteristics

Yields. This paper uses one yield per asset class as its price. For Treasuries I use a representative constant-quality series that spans the full sample, the 10-year constant-maturity Treasury yield (DGS10) from FRED. A point yield at a representative maturity tracks the long end of the market, where habitat demand bites, more cleanly than a whole-market index does over a four-decade sample. For agency MBS I use an option-adjusted yield, the 10-year Treasury yield plus the option-adjusted spread of the Bloomberg agency MBS index (available from 1990Q3, which sets the start of the MBS estimation sample): the index's own yield-to-worst embeds the prepayment-option premium, which moves with interest-rate volatility rather than with the compensation investors require for holding MBS. Table A.2 in the Appendix reports the sources and ticker codes. I aggregate all series to the quarterly frequency, taking the end-of-quarter value.

Other Characteristics. The system uses four asset characteristics: maturity, net issuance, duration, and coupon. The weighted time-to-maturity, modified duration, and weighted-average coupon of each index come from Bloomberg. Duration enters lagged one quarter, because its change is a near-mechanical function of the contemporaneous yield change ($dD \approx -D\, dy/(1+y)$). The coupon is a slow-moving composition variable and is the exogenous shifter behind the Treasury heteroskedasticity instrument (Section 3.6). Net issuance is measured from the Z.1 flow tables, as every holder's transactions in the asset, the Federal Reserve included, over the lagged amount outstanding, rather than from the change in market-value outstanding, which is mostly revaluation and would enter the estimation as a function of the yield itself. Table A.2 in the Appendix lists the sources for each asset class.

Macroeconomic Variables. I include a set of macroeconomic controls to absorb time-series variation in aggregate portfolio demand unrelated to asset prices. Specifically, I control for log real GDP, real GDP growth, core inflation, and the log of the U.S. dollar index, all obtained from the Federal Reserve Economic Data (FRED) database and aggregated to the quarterly frequency. I also include the one-quarter lag of log VIX as a measure of aggregate risk sentiment. Because VIX is jointly determined with asset prices during flight-to-quality episodes, I use the lagged value as a predetermined control to avoid simultaneity bias. The estimation also includes an indicator for the 12 crisis quarters in which the instruments are switched off (Section 3.6).

2.3 Instruments

The main instruments come from two sources, as Section 3.6 explains: the unexpected part of the Fed's asset purchases, for both markets the Fed trades; and a crisis-robust heteroskedasticity-based instrument built from the coupon characteristic already in the panel, which the Treasury block uses alongside it. A granular instrument built from the sectors' holdings, maturity walls, and the monetary-policy surprises of the prior demand-

system literature enter as diagnostics only. This subsection records the data sources.

Unexpected Federal Reserve Asset Purchases. The instrument for both markets is the unexpected part of the Fed's asset purchases. Realized purchases are not a surprise, because once a program is announced they follow a published monthly schedule; the surprise is the revision to that schedule. For each of the 21 FOMC communications between November 2008 and May 2024 that set or restated the announced path of System Open Market Account purchases, I record the revision to expected net purchases of agency MBS and of Treasuries over the following four quarters, relative to the previously announced path, date it to the quarter in which the change first became public, and scale it by the amount outstanding. Appendix D.4 describes the construction, and Appendix Table A.5 lists every event.

2.4 Summary Statistics

Table 2 reports summary statistics for the merged dataset. Panel (a) presents average portfolio shares and assets under management (AUM) by sector type. Foreign investors allocate the largest share of their portfolios to Treasuries, 28.9 percent, and arbitrageurs and banks the largest to agency MBS, 11.5 and 10.8 percent. Households and pension funds hold small fixed-income shares despite their large aggregate AUM. Most of each market's float sits with sectors that, Section 4 shows, do not trade on yields: in 2025Q1 foreign investors, banks, pension funds, and insurers together hold 56 percent of the private coupon float, and the two sectors whose active demand does respond to Treasury yields, the household residual and the mutual funds, hold about a seventh each. This concentration is what makes the slope of the Treasury market's aggregate demand curve small. Panel (b) reports the characteristics of the two assets. Their yields average almost the same, 4.8 percent for Treasuries and 4.7 for agency MBS, but the Treasury index is the longer one, 8.1 years of maturity against 6.8 and 5.5 years of duration against 4.8, prepayment compressing the effective duration of MBS. The estimation sample covers 27 sectors observed quarterly from 1985Q1 to 2025Q1.

Table 2: **Summary Statistics**

(a): Portfolio Summary

	Financial Assets ($T)		Portfolio Weight (%)	
Sector Type	Total	Bond	Treasury	Agency MBS
Banks	12.87	1.91	3.60	10.78
Insurers	6.24	0.59	4.42	7.08
Pension Funds	13.21	0.74	4.29	3.05
Mutual Funds	12.07	1.41	5.67	7.13
Arbitrageurs	3.22	0.45	3.39	11.50
Foreign	19.68	6.62	28.94	4.85
Household (incl. hedge funds)	50.04	0.79	1.09	0.49
Other	30.91	1.52	2.23	3.72

(b): Asset Summary

Asset	Variable	Mean	Median	S.D.	Min	Max	N
Treasury	Yield (%)	4.77	4.44	2.39	0.66	11.65	161
	Maturity (years)	8.06	8.16	0.94	6.08	9.88	161
	Duration (years)	5.49	5.35	0.63	3.93	7.11	161
	Coupon (%)	5.29	4.97	2.83	1.57	11.55	161
Agency MBS	Yield (%)	4.71	4.54	2.03	1.31	9.60	139
	Maturity (years)	6.75	6.78	1.63	2.15	12.31	161
	Duration (years)	4.76	4.84	0.84	1.84	6.38	161
	Coupon (%)	6.06	5.50	2.32	2.60	10.73	161

(c): Instruments

Asset	Instrument	Mean	Median	S.D.	Min	Max	N
Treasury	Fed path revision (%)	0.14	0.00	1.26	-3.94	12.17	159
	Coupon heteroskedasticity	0.00	0.00	0.03	-0.11	0.23	159
Agency MBS	Fed path revision (%)	0.07	0.00	1.05	-3.48	9.52	138

Notes: This table shows summary statistics for the estimation sample, from the Z.1 Financial Accounts. Panel (a): sector financial assets and time-series average portfolio weights, with the Treasury weight on the coupon market. Panel (b): asset–quarter statistics; N is the number of quarters, and the agency MBS yield starts in 1990Q3. Panel (c): the instruments as they enter the second stage, the Fed path revision in percent of the previous quarter's amount outstanding (Appendix D.4) and the coupon instrument, the demeaned coupon change times the first-step residual (Section 3.6); both are zero in the 12 crisis quarters.

3 Empirical Framework

This section develops a demand system for the two U.S. bond markets the Federal Reserve trades, Treasury securities and agency MBS, identified from sector-level portfolio data. Four features set it apart from existing bond demand systems. First, demand is measured from transactions rather than from the change in the market value of holdings, so the yield coefficient is identified from trading. Second, the outside asset differs across sectors: I measure each sector's bond demand against the non-bond class it actually rebalances toward, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers (e.g., Rodnyansky and Darmouni, 2017; Kandrac and Schlusche, 2021; Kurtzman et al., 2022). Table 1 lists the outside asset for each sector and Appendix D.1 gives the rationale. Third, each sector's yield sensitivity differs by asset class, so the system can say who absorbs Treasuries and who absorbs agency MBS instead of imposing one slope per investor across both markets. Fourth, demand adjusts at a sector-specific speed, estimated horizon by horizon with local projections, which separates the impact slope from the long-run slope the counterfactual uses.

The sector-specific outside asset has a cost and a benefit. The cost is that the slopes are not directly comparable across sectors, since each is defined relative to a different outside asset; the aggregate demand slope of Section 3.4, the object the counterfactual requires, aggregates across sectors whatever outside asset each uses. The benefit is that the substitution margin is the one the portfolio-balance channel of QE and QT operates on. Jansen et al. (2024), Fang et al. (2025), and Eren et al. (2026) measure demand against other bonds, and investors substitute far more readily between one bond and another than between bonds and equities or loans, so a within-bond outside asset yields a more elastic demand than the bond-versus-non-bond margin that governs the price impact of central-bank purchases.

The use of aggregate shares is analogous to the market-level setup of Berry et al. (1995), without random coefficients, because all sectors face the same prices and characteristics at each date; Appendix A gives the mapping.

3.1 Setup

I index sector-level investors by $i = 1, \ldots, I$; sectors include, for example, life insurers and mutual funds. Assets are indexed by $n = 0, \ldots, N$, where $n = 0$ denotes the outside asset, which differs across sectors and comprises non-bond holdings such as equities and loans. The bond universe consists of Treasury securities and agency MBS, the two markets the Federal Reserve trades. Corporate and municipal bonds are not in it: no instrument for their yields survives the measurement discipline of Section 3.6, so their demand slopes cannot be estimated. They are not folded into the outside asset either: they sit outside the choice set. In the logit that leaves the estimates unchanged, because the log ratio of a bond's share to the outside asset's share does not depend on which other alternatives the investor holds. Appendix D.1 uses them as a common outside asset instead. The yield of asset n in quarter t is $y_t(n)$. Each asset has a vector of observed characteristics $x_t(n)$: maturity, net issuance measured from transactions, and duration. The characteristics enter the estimation predetermined; in particular, duration enters lagged one quarter, because the change in modified duration is a near-mechanical function of the contemporaneous yield change (Section 2). The portfolio weight of investor i in asset n at time t is $w_{it}(n)$, with $\sum_{n=0}^{N} w_{it}(n) = 1$.

3.2 Portfolio Choice

I assume that portfolio shares follow a characteristics-based logit demand, under which the share is exponential in a linear index of characteristics (Koijen and Yogo, 2019).[3] The portfolio weight $w_{it}(n)$ is

$$w_{it}(n) = \frac{\delta_{it}(n)}{1 + \sum_{k=1}^{N} \delta_{it}(k)}, \tag{5}$$

where

$$\delta_{it}(n) = \exp\Big\{\beta_i(n)\, y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n)\Big\}\epsilon_{it}(n). \tag{6}$$

[3] Koijen and Yogo (2019) provide an asset-pricing-based theoretical foundation for logit demand by deriving it from mean–variance portfolio choice under factor-structure assumptions.

$\beta_i(n)$ is investor i's demand semi-elasticity with respect to the yield of asset n, and γ_i (a vector) collects investor i's loadings on asset characteristics. $\xi_{it}(n)$ denotes an unobserved demand shifter capturing asset–time variation not explained by observed characteristics, and $\epsilon_{it}(n)$ is the latent demand not explained by the observed yield or characteristics. The outside-asset weight is

$$w_{it}(0) = \frac{1}{1 + \sum_{n=1}^{N} \delta_{it}(n)}. \tag{7}$$

The yield slope carries an asset index. Existing bond demand systems impose $\beta_i(n) = \beta_i$: a one-percentage-point rise in the yield of any bond shifts investor i's log-odds of holding it by the same amount. I let the slope differ by asset, the alternative-specific-coefficient logit familiar from discrete choice. The share structure is unchanged: substitution across assets still follows the logit's proportional pattern, and the outside asset still absorbs the balance, so the own- and cross-yield responses of Section 3.4 keep their closed forms. What the generalization drops is only the equal-slope restriction, and Section 4 shows that the data reject it: mutual funds, for example, absorb Treasuries but not the mortgage spread, and banks absorb agency MBS but not Treasuries. The common-slope system is the special case $\beta_i(n) = \beta_i$.

The price that enters each block, $y_t(n)$, is the one that market itself sets: the 10-year yield for Treasuries and, for agency MBS, the option-adjusted spread over it, since the MBS yield is the Treasury yield plus that spread and the Treasury leg carries eight times its variance at the quarterly frequency. Writing MBS demand on the yield level instead would impose that a sector's share responds equally to the Treasury yield and to the spread; Section 4 tests the two restrictions against each other; the test supports the spread specification without settling it, and the choice is a modeling decision under which a shift in MBS supply reaches the Treasury level only through the logit shares, a small channel. With this choice the clearing system of Section 5 is nearly block-recursive: the Treasury market sets the level, the agency MBS market sets the spread, the two interact only through the logit shares, and the mortgage-relevant yield is their sum.

Combining (5) and (7) yields the estimable specification:

$$\begin{aligned}\ln \frac{w_{it}(n)}{w_{it}(0)} &= \ln \delta_{it}(n) \\ &= \beta_i(n)\, y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n) + \varepsilon_{it}(n). \end{aligned} \tag{8}$$

The left-hand side is the log portfolio share of asset n relative to the outside asset. The right-hand side is a linear combination of the asset's yield $y_t(n)$, characteristics $x_t(n)$, and asset–time demand shifter $\xi_{it}(n)$ with estimable coefficients; $\varepsilon_{it}(n)$ is an idiosyncratic demand shock, the log of the latent demand $\epsilon_{it}(n)$. I parameterize the asset–time demand shifter as

$$\xi_{it}(n) = \alpha_n + \nu_{in}' W_t + \tilde{\xi}_{it}(n),$$

where W_t is a vector of macroeconomic controls, ν_{in} captures sector- and asset-specific loadings on macro conditions, and $\tilde{\xi}_{it}(n)$ is the residual unobserved demand shifter. I follow Jansen et al. (2024) and Eren et al. (2026) in parameterizing the asset–time demand shifter via macro controls rather than asset–time fixed effects, which would absorb the time-series variation in the instruments.

I estimate the demand equation in first differences. Differencing (8) is an algebraic transformation that leaves every slope unchanged,

$$\Delta \ln \frac{w_{it}(n)}{w_{it}(0)} = \beta_i(n)\, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \Delta \varepsilon_{it}(n). \tag{9}$$

In estimation the left-hand side is measured net of the revaluation of positions already held, by $\Delta q_{it}(n)$ of Equation (4). Differencing has two benefits. First, it removes the asset fixed effect α_n. Second, it makes both sides of the equation stationary.

3.3 Rebalancing Speed

Equation (9) recovers the semi-elasticity $\beta_i(n)$ only if sectors complete their rebalancing within the quarter, and they do not: asset–liability mandates, illiquid positions, and the slow-moving capital of Duffie (2010), which Jiang and Sun (2024) model as a portfolio adjustment cost for long-term bond investors, spread the response over several quarters. Estimated on one-quarter changes, (9) delivers only the *impact* response. I therefore treat (8) as the portfolio a sector ultimately wants and let observed positions converge to it at a sector-specific speed, as van der Beck (2026) do for equity institutions, and trace the adjustment with instrumental-variables local projections (Jordà, 2005):

$$\Delta^{(h)} \ln \frac{w_{i,t+h}(n)}{w_{i,t+h}(0)} = \beta_i^{(h)}(n)\, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \tau_{in}\, t + u_{it}^{(h)}(n), \tag{10}$$

where $\Delta^{(h)}$ takes the change from quarter $t-1$ to quarter $t+h$, measured as the cumulative transaction growth $\sum_{s=t}^{t+h} \Delta q_{is}(n)$; $\Delta y_t(n)$ is instrumented as in Section 3.6; the linear trend $\tau_{in}\, t$ absorbs the secular comovement of the long sample; and standard errors are Newey–West with bandwidth h (Newey and West, 1987). Only the response window extends with h; the yield change stays a one-quarter change, so the specification does not reintroduce the persistence that differencing removes. At $h = 0$, (10) is (9) and $\beta_i^{(0)}(n)$ is the impact slope; as h grows, $\beta_i^{(h)}(n)$ rises toward the full response. I estimate (10) for each sector type and asset, pooling the member sectors of the type.

I read the long-run slope one year out, at $h = 4$, a horizon fixed in advance, and take it as the semi-elasticity $\beta_i(n)$ of (8): whatever the frictions, once rebalancing has run its course the allocation is the one the static system describes. For absorbers that are still adjusting after a year, the one-year slope is a lower bound on that response. The speed of adjustment is a friction, not a demand parameter, so the aggregate demand slope and the counterfactual equilibrium use $\beta_i(n)$, and the transition to that equilibrium uses the path $\beta_i^{(h)}(n)$.

3.4 Demand Elasticity and the Aggregate Demand Slope

The logit demand system yields a closed-form expression for the yield elasticity of demand. The elasticity of the portfolio share of asset n with respect to the yield of asset m is

$$\eta_{it}(m,n) = \frac{\partial \ln w_{it}(n)}{\partial \ln y_t(m)} = \begin{cases} \beta_i(n) \cdot \big(1 - w_{it}(n)\big) \cdot y_t(n) & m = n \\ -\beta_i(m) \cdot w_{it}(m) \cdot y_t(m) & m \neq n, \end{cases} \tag{11}$$

where $\beta_i(n)$ is the long-run yield sensitivity of sector i for asset n from Section 3.3, $w_{it}(n)$ is the portfolio share of asset n, and $y_t(n)$ is the yield of asset n in quarter t. The own elasticity ($m = n$) gives the percentage change in the portfolio share per one-percent rise in the asset's own yield; the cross elasticity ($m \neq n$) is the spillover to other assets, which scales with the shocked asset's slope and portfolio share.

The object that governs the yield effect of a supply shift is not the elasticity but a yield-free slope. Linearizing the market-clearing condition of the next subsection around the observed equilibrium, a supply shift $\Delta S_t(n)$ must be absorbed by the sectors' share adjustments, $\sum_i A_{it}\, \beta_i(n)\, w_{it}(n)\big(1 - w_{it}(n)\big)\, \Delta y_t(n) = \Delta S_t(n)$, so removing 1 percent of asset n's private float raises its yield by

$$\Delta y_t(n) = \frac{1}{\sum_i s_{it}(n)\, \beta_i(n)\, \big(1 - w_{it}(n)\big)} \text{ basis points}, \tag{12}$$

where $s_{it}(n) = A_{it}\, w_{it}(n) / S_t(n)$ is sector i's share of the float and A_{it} its assets under management. The denominator is the slope of the aggregate demand curve for market n, which I call the aggregate demand slope. It aggregates the holders' own-asset sensitivities weighted by their share of the float: a market whose float sits with sectors that do not trade on yields has a small slope and a large price impact, whatever those sectors' behavior in other markets. Investors respond to yields point for point, so the yield level cancels and the basis-point impact is pinned down by this slope alone; its movement over time reflects the composition of holdings. Equation (12) is the first-order version of the counterfactual in Section 5, which

clears the full nonlinear system.

3.5 Market Clearing

To close the asset pricing model, the market for each asset n clears in each period t:

$$\sum_i A_{it} \cdot w_{it}(n) = S_t(n), \tag{13}$$

where A_{it} is investor i's total wealth at time t and $S_t(n)$ denotes the outstanding supply of asset n at time t. Market clearing holds in this study because, by construction, the aggregate share data cover all investors and sectors. The counterfactual imposes it in both markets.

3.6 Identification

The yield is endogenous: sector-level demand shocks move prices, the more so in a setting that aggregates investors into sectors rather than assuming atomistic behavior. I instrument each asset's yield with an instrument specific to that asset, and require that its variation move portfolio shares only through that yield, after macroeconomic controls. Because the estimating equations are the differenced specification (9) and the local projections (10), whose horizon-h error cumulates the latent-demand revisions from t through $t+h$, the restriction applies over the projection window:

$$\mathbb{E}\big[\Delta\varepsilon_{i,t+s}(n) \mid Z_t(n)\big] = 0, \qquad s = 0, 1, \ldots, h. \tag{14}$$

At $h = 0$ this is the standard contemporaneous exclusion restriction; at longer horizons it adds the lead-exogeneity condition of LP-IV (Stock and Watson, 2018), that the instrument be unpredictable not only for current latent demand but for its revisions over the year the response cumulates.

Treasury Yields. No external supply shifter is available for Treasury yields over four decades: the Fed's own transactions track the yield cycle, maturity walls (Almeida et al., 2012; Harford et al., 2014) are predetermined but too small to be relevant, realized issuance responds to rates, and foreign demand chases yields. I therefore identify the Treasury slopes with the heteroskedasticity-based instrument of Lewbel (2012). At the asset–quarter level I regress the yield change on the differenced characteristics, take the residual $\hat{e}_t(n)$, and interact it with the demeaned coupon,

$$Z_t(n) = \left[\Delta x_t(n) - \overline{\Delta x(n)}\right] \hat{e}_t(n). \tag{15}$$

The instrument is valid if the characteristic is uncorrelated with the product of the demand and yield shocks, and relevant if the variance of the yield shock moves with the characteristic. The two conditions pull against each other in bond markets: yield-shock variance is highest in crisis quarters, and in exactly those quarters flight to quality makes demand and yield shocks covary. I therefore identify from calm-period heteroskedasticity alone. The crisis quarters are those of the flight-to-quality episodes documented in the literature, the 1987 crash (1987Q4), the LTCM collapse (1998Q3), September 11 (2001Q3), the 2008 panic (2008Q3–Q4), the U.S. downgrade and euro-area crisis (2011Q3), the March 2020 dash for cash (2020Q1–Q2), and the 2022 tightening (2022Q1–Q4); I set the instrument to zero in these 12 quarters and control for them in the second stage. Alongside it I use the Treasury leg of the purchase-path revisions described next, dated to the quarter in which each change first became public rather than to the statement that formalizes it, since the Committee signals a change before it votes on it. Neither instrument is strong, so a weak-instrument-robust Anderson–Rubin set (Anderson and Rubin, 1949; Andrews et al., 2019) accompanies every slope.

Agency MBS Yields. The Fed is the one buyer of agency MBS whose purchases are not a response to any private sector's demand, but its *realized* purchases follow a monthly schedule announced in advance and carry no surprise. The information is in the revisions to that

schedule. For each of the 21 FOMC communications between November 2008 and May 2024 that set or restated the announced path of System Open Market Account purchases, I record the implied revision to expected net purchases over the following four quarters, separately for agency MBS and for Treasuries and scaled by the outstanding stock. An asset-specific supply shifter has to move the price of its own market and leave the other's untouched, and Section 4 reports that placebo, along with the same test applied to the alternatives. The instrument is set to zero in the same 12 crisis quarters as the heteroskedasticity instrument, where the announcement and the flight to quality are inseparable. Appendix D.4 describes the construction and lists the events, and Appendix D.5 reports the menu of alternatives and the checks that the revisions do not respond to the previous quarter's yields or macroeconomic conditions.

3.7 Non-Logit Demand Sectors

The demand system asks each sector to choose between a bond and an outside asset at the bond's yield. Two sectors do not fit that description, and both of them hold Treasuries.

Money market funds hold bills. Rule 2a-7 caps a money fund's weighted average maturity at 60 days and each holding at 397 days, so the 10-year yield is not the price its Treasury demand responds to. Bills are 78 percent of money funds' Treasury holdings today and 51 percent on average since 1974, against 0.7 percent for the long-duration mutual funds with which the Financial Accounts group them. I therefore estimate the mutual-fund type without money funds and treat the money-fund cell as inelastic in the clearing: their holdings stay in the float and their response is zero.

Dealers are intermediaries. A dealer's Treasury book is inventory financed by repo, not a portfolio choice: when yields rise the book loses value and a levered intermediary sells to

restore its balance sheet. An intermediary holding leverage constant would sell

$$\frac{\partial \ln H_t(\text{dealer, Treasury})}{\partial y_t} \;=\; -\,\lambda_t\, D_t\, s_t^{\text{bond}}/100 \;=:\; \text{rule}_t, \tag{16}$$

where λ_t is leverage from the Financial Accounts, D_t the index duration, and s_t^{bond} the bond share of financial assets, all lagged one quarter so that rule_t is predetermined. I replace the dealer cell's free slope with $\varphi\,\text{rule}_t$ and estimate the single parameter φ on the same instruments, using $Z_t\,\text{rule}_t$ as the instrument for $\text{rule}_t\,\Delta y_t$, so the cell's contribution moves with the observed balance sheet. The rule is applied to the Treasury cell only; Section 4 reports φ in both markets.

The estimation sample is part of the identification. The household residual's Treasury slope is estimated on the 1985–2025 span, because the 2010–2025 window alone does not identify it (Section 4 and Appendix D.3), so the Treasury counterfactuals rest on the pre-2010 behavior of that sector. The agency MBS absorbers are identified in both windows.

3.8 Scope of the Demand-System Approach

The answers here are valid within the known limits of the demand-system approach (Fuchs et al., 2023; Koijen and Yogo, 2025). First, a logit demand system need not correspond to a valid stochastic discount factor or rule out arbitrage (Fuchs et al., 2023; Kocherlakota, 2025); I use it as a description of how holders trade, not as a model of prices derived from preferences. Second, the logit restricts substitution: letting the slope differ by asset relaxes the restriction within each holder, but the cross-asset responses still follow from the portfolio shares rather than being estimated separately, and the counterfactuals inherit that assumption. Third, identifying a demand system requires as many independent sources of price variation as priced assets (Fuchs et al., 2025). With two asset classes and an instrument set for each, the system meets that requirement, and because the instruments are aggregate time-series shocks, the slopes carry the aggregate response of demand rather than only its

cross-section (Haddad et al., 2025). The requirement is also one of direction: a supply shock identifies an own-price slope only if the two assets do not overlap in the states where they pay. Measuring the MBS price as the spread over Treasuries separates them along that margin, which the cross-asset placebo of Section 4 verifies at the observed equilibrium. Finally, the slopes are identified from quarterly variation around the observed equilibrium, while the largest programs move supply far outside it, so the counterfactuals are most reliable per unit of supply and least reliable for the largest episodes. The agency MBS counterfactuals also hold the prepayment-option component of the MBS yield fixed: a program that changes interest-rate volatility or refinancing behavior moves mortgage rates through that component as well, a channel outside the demand system.

4 Results

4.1 First Stage

Table 3 reports the first stage on the 1985–2025 asset–quarter panel, with the same controls as the second stage and Newey–West standard errors. Each market the Fed trades is identified by instruments that carry the sign theory predicts: two for Treasuries, one for agency MBS. The Treasury block pairs the coupon interaction ($t = -3.7$) with the revision to the Fed's announced Treasury purchase path ($t = -1.8$) for a joint F of 7.7, and the overidentifying restriction survives in every sector type. The agency MBS block is the stronger of the two: an upward revision to the announced MBS purchase path compresses the option-adjusted spread with an F of 18.1 ($t = -4.3$). The Fed's realized transactions reach only 4.2, and 3.1 when paired with the Drukker (2025) surprise, because the realized flow follows the schedule the announcement already set. The revision passes the cross-asset placebo an asset-specific supply shifter has to pass, leaving the Treasury yield at an F of 1.0, and no alternative shifter I examined passes it with a usable first stage (Appendix Table A.7). Because the yield varies only at the asset–quarter level, the first stage is common to every sector, so

differences in the estimated slopes reflect demand heterogeneity rather than differences in the identifying variation. I read every slope through its weak-instrument-robust Anderson–Rubin set and call a slope different from zero only when that set excludes zero. Any remaining bias has a known direction: weak instruments pull two-stage least squares toward ordinary least squares, which the simultaneity of demand pushes down, so the point estimates are if anything lower bounds.

Table 3: **First Stage**

Asset	Instrument	Coefficient	First-stage F	N
Treasury	Lewbel (coupon)	-4.346^{***}	7.7	159
		(-3.73)		
	Fed Treasury purchase-path revision	-1.750^{*}		
		(-1.82)		
Agency MBS	Fed MBS purchase-path revision	-5.746^{***}	18.1	138
		(-4.25)		
Joint F (all instruments)			8.0	297

Notes: This table shows the first stage of the yield change on the instruments, asset by asset, on 1985Q1–2025Q1 quarterly data (agency MBS spread from 1990Q3), with the second-stage controls and Newey–West t-statistics (four lags) in parentheses. The instruments are defined in Section 3.6 and set to zero in the 12 crisis quarters. The first-stage F is the Newey–West Wald statistic of each asset's own instruments; the joint F is for the stacked asset–quarter panel with asset fixed effects.

Two measurement choices carry the identification. For Treasuries the market-value granular instrument reaches a first stage of 10.4 and looks like the strongest instrument available; built from the donors' transactions it reaches 4.3, so most of its relevance was the revaluation that also violates its exclusion restriction, and I drop it. For agency MBS, the Fed's realized transactions, which follow the announced schedule, reach a first stage of only 4.2 on the spread, and replacing the realized flow with the revisions to the announced purchase path raises it to 18.1: the realized flow is anticipated and the revisions are not.

Three further checks bear on the identification (Appendix Table A.6). First, a three-donor Treasury block leaves the long-run slopes essentially unchanged at a first-stage F of 6.4, so the donor choice affects precision rather than the point estimates. Second, the agency MBS

block does not use heteroskedasticity instruments: adding the Lewbel donors to the Fed's realized transactions and the Drukker surprise raises that set's first stage from 3.1 to 17.3, but the overidentification test then rejects for insurers, mutual funds, and the arbitrageurs at $p \leq 0.05$, because the added relevance comes from the maturity interaction, and the index maturity of agency MBS moves with prepayment and hence with the yield. I keep the single purchase-path instrument, just-identified, and read its slopes through their Anderson–Rubin sets. Third, LIML and Fuller estimates coincide with two-stage least squares in every cell, so the weak-instrument bias of the point estimates is not the issue; the width of the sets is.

4.2 Demand Slopes by Sector

Table 4 reports, for each sector type and each of the two markets the Fed trades, the impact slope $\beta_i^{(0)}(n)$, the long-run slope $\beta_i^{(4)}(n)$, and the 95 percent Anderson–Rubin set for the long-run slope. I take the long-run slope as the cumulative response one year after the shock, a horizon fixed in advance by which most of each absorber's response has arrived; Figure 2 shows the paths to two years. Each market has its own, thin set of active absorbers, and the set differs across markets.

In the Treasury market the marginal buyer is the household residual, whose long-run slope is 0.73: a one-percentage-point rise in the 10-year yield raises its Treasury position relative to its equity holdings by about 73 percent in log terms. Mutual funds are second at 0.28, once money market funds are taken out of that sector type (Section 3.7); the money-fund cell has a slope of -0.56 against the 10-year yield, and pooling it in removes the type's slope, which falls to 0.02. Banks are third at 0.09, and the rest have point estimates near zero: pension funds at 0.03, insurers at -0.04, and foreign investors, the largest holders, at -0.00. Only the first two Anderson–Rubin sets exclude zero. The near-zero slopes are failures to reject rather than precise zeros: the sets reach 0.36 for banks and 0.12 for foreign investors, and under the external instruments of Appendix D.5 banks absorb at 0.51 and foreign investors at 0.33, so the level of the aggregate slope is better determined than the

identity of the marginal buyer. The arbitrageur cell, dealers and mortgage REITs, carries a large negative slope, -1.67, which is the momentum of an inventory and basis book rather than a demand curve; the balance-sheet rule of Equation (16) replaces it, with $\varphi = 4.12$ (standard error 2.53), procyclical leverage in the sense of Adrian and Shin (2010), implying a slope of -0.71 at the 2025Q1 balance sheet. On agency MBS the same regression returns $\varphi = -4.16$, the opposite of deleveraging, so the rule is applied to the Treasury cell only. The foreign type pools two units that offset each other: reserve managers are flat, at 0.06, and private foreign investors chase returns, at -0.06.

The household residual is what the Financial Accounts assign to no measured sector, and after 2010 that includes the hedge funds whose cash–futures basis positions dominate its Treasury flows (Jansen et al., 2024). The Enhanced Financial Accounts show how far this has gone: hedge funds hold 122 percent of the residual's coupon Treasuries on average over 2013–2025, and the residual's Treasury transactions load on the quarterly change in hedge-fund holdings with a coefficient of 0.62 (Appendix D.6). The two are the same position, which matters because the slope is identified where the residual was households: the 2010–2025 window on its own gives -0.84 with a standard error of 0.82, an interval that spans every value the counterfactual could use. Whether a levered arbitrageur absorbs supply the way a buy-and-hold household did is the open question behind the Treasury band of Section 5. The cell is also the one most exposed to the measurement of transaction growth, because the residual itself is near zero or negative between 2004 and 2008 (Appendix D.2). And because the residual is total outstanding less every identified holder, it absorbs the measurement error in every other sector's recorded transactions, so its slope may partly reflect yield-correlated error in those series; the counterfactual inherits that risk.

In the agency MBS market, where the price is the spread, every sector type slopes the right way. The household residual leads at 2.33, followed by foreign investors at 0.50, banks at 0.48, and pension funds at 0.26; mutual funds, the arbitrageurs, and insurers are positive but imprecise. Four of the eight sets exclude zero, against two in the Treasury market. That

every slope is positive is not imposed: with the MBS yield level as the price, banks, pension funds, and the residual sectors carry negative long-run slopes, which is what a demand curve should never do. The data support the choice of price, without settling it. Entering the Treasury yield and the spread separately, each with its own instrument, the level coefficient is not different from zero at the 5 percent level in any sector type, while the equality a yield specification imposes is rejected for pension funds ($p = 0.03$) and foreign investors ($p = 0.05$). Foreign absorption runs through both units, the official sector at 0.68 and the private sector at 0.32. In the 2010–2025 window the household residual, the arbitrageurs, banks, and pension funds keep their signs, while mutual funds turn negative and insurers and foreign investors sit at zero.

The data also reject the equal-slope restriction of the standard system, at the 10 percent level. A minimum-distance test of the restriction that each type's Treasury and agency MBS slopes are equal gives 15.1 on 8 degrees of freedom ($p = 0.06$), driven by foreign investors, pension funds, and the household residual (Appendix Table A.10). An intermediate structure, $\beta_i(n) = \beta_i\,\theta(n)$, is not rejected ($p = 0.29$), with loadings of 0.32 on the Treasury yield against 1.38 on the agency MBS spread: the same investor responds four times as strongly to a basis point of spread as to a basis point of level. The counterfactual uses the unrestricted slopes.

4.3 Rebalancing Speed

Figure 2 traces the horizon response $\beta_i^{(h)}(n)$ of every sector type in both markets, two years out. Every absorber builds its position slowly, and in the agency MBS market the build is monotone quarter by quarter. Banks move from 0.12 on impact to 0.25, 0.41, and 0.48 at one year; foreign investors from 0.09 to 0.24, 0.35, and 0.50; insurers from 0.02 to 0.04. The Treasury absorbers do the same, the household residual moving from 0.10 on impact to 0.73 at one year and mutual funds from 0.07 to 0.28. For these cells the impact response is between an eighth and a quarter of the one-year response, and pension funds are the one

Table 4: **Demand Slopes by Sector Type and Asset**

	Treasury (yield)		Agency MBS (spread)	
Sector type	$\beta^{(0)}$	$\beta^{(4)}$	$\beta^{(0)}$	$\beta^{(4)}$
Banks	+0.02	+0.09	+0.12	+0.48
	(0.04)	(0.11)	(0.06)	(0.25)
	[-0.06, 0.36]		[0.03, 0.98]	
Insurers	+0.01	-0.04	+0.02	+0.04
	(0.03)	(0.06)	(0.02)	(0.11)
	[-0.16, 0.14]		[-0.18, 0.24]	
Pension Funds	+0.03	+0.03	+0.16	+0.26
	(0.02)	(0.04)	(0.06)	(0.13)
	[-0.04, 0.11]		[0.04, 0.54]	
Mutual Funds	+0.07	+0.28	+0.06	+0.16
	(0.04)	(0.12)	(0.11)	(0.30)
	[0.05, 0.65]		[-0.37, 0.81]	
Arbitrageurs	+0.20	-1.67	-0.11	+0.15
	(0.36)	(1.14)	(0.22)	(0.41)
	[-8.52, 0.16]		[-0.76, 0.90]	
Foreign	-0.04	-0.00	+0.09	+0.50
	(0.02)	(0.06)	(0.06)	(0.24)
	[-0.10, 0.12]		[0.12, 1.12]	
Household (incl. hedge funds)	+0.10	+0.73	-1.04	+2.33
	(0.12)	(0.28)	(0.63)	(0.92)
	[0.18, 1.91]		[1.04, 5.10]	
Other	+0.08	+0.04	+0.04	+0.20
	(0.07)	(0.14)	(0.19)	(0.44)
	[-0.26, 0.34]		[-0.68, 1.08]	
N	3,259		2,864	

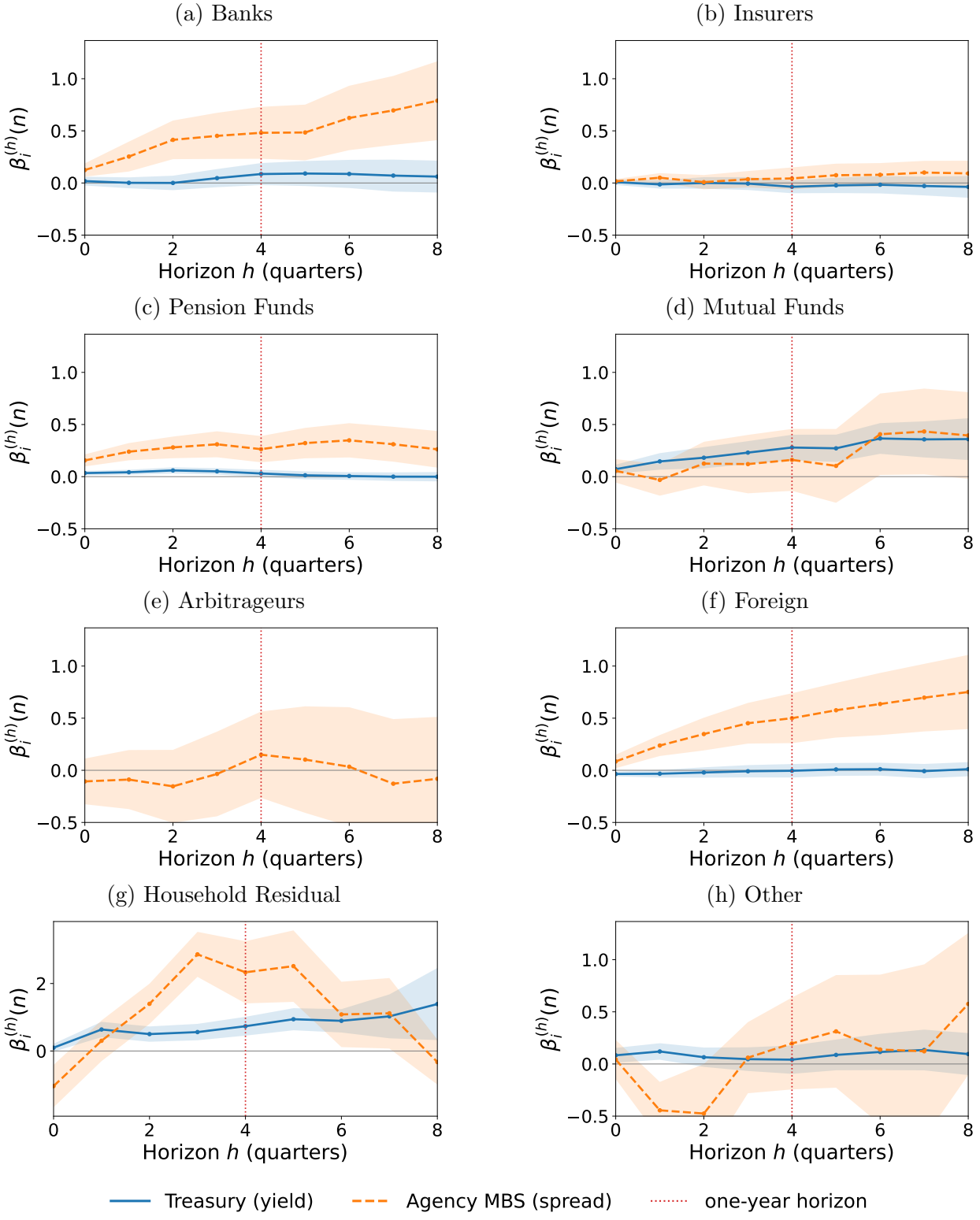
Notes: This table shows the sector-type slopes from the local projections of Equation (10), pooled over each type's member sectors, on 1985Q1–2025Q1 (agency MBS from 1990Q3), with the instruments of Table 3. $\beta^{(0)}$ is the impact slope and $\beta^{(4)}$ the cumulative one-year response, per percentage point of the Treasury yield or of the agency MBS spread; Newey–West standard errors in parentheses, 95 percent Anderson–Rubin sets for $\beta^{(4)}$ in brackets (^{u}: set reaching the grid edge; $\emptyset$: empty set). N is sector–quarter observations. The two cells of Section 3.7, money market funds and the dealer Treasury book, follow their own models in the counterfactual.

fast absorber, at 0.16 of 0.26 within the quarter. The slow-moving capital of Duffie (2010) is thus a property of every sector that absorbs supply, not of a particular investor class, and it operates at the scale of quarters. Several absorbers keep building beyond the first year: banks' and foreign investors' agency MBS slopes and the mutual funds' Treasury slope rise by a third to two-thirds more by $h = 8$, although the estimates there are less precise. The one-year slope the counterfactual uses therefore understates their full response, so if anything the counterfactual overstates the long-run yield effects.

Two cells trade the other way in the quarter of the shock and reverse afterwards. The household residual's agency MBS slope is -1.04 on impact and $+2.33$ at one year, and the arbitrageurs' is -0.11 and $+0.15$; both are momentum in the shock quarter and rebalancing after it. Their weight is large enough to matter for the market: at the 2025Q1 composition the agency MBS aggregate demand slope is slightly *negative* on impact and turns positive from the first quarter on. The MBS market has no marginal buyer in the quarter a supply shock lands, and acquires one a quarter later. Section 5 reports the consequence, which is that the impact-quarter equilibrium of a supply shift is not defined and the yield path of a runoff becomes defined only from the following quarter.

A static quarterly regression would therefore understate the absorbers' demand four- to eightfold. The distinction matters for the counterfactual: the impact response governs the initial dislocation, the long-run slope where yields settle, and the gap between them is the overshoot a balance-sheet shock produces before slow capital arrives.

Figure 2: **Rebalancing Paths by Sector Type**



Notes: This figure shows the horizon responses $\beta_i^{(h)}(n)$ of Equation (10) with ±1 Newey–West standard error bands, 1985Q1–2025Q1 (agency MBS from 1990Q3); the dotted line marks the one-year horizon. Treasury and spread slopes are in different units. The arbitrageur Treasury cell follows the leverage rule of Section 3.7 and is omitted.

4.4 The Aggregate Demand Slope

The slopes and the composition of holdings together determine how much a market's yield moves per unit of supply. Figure 3 plots the implied price impact of Equation (12), the inverse of the aggregate demand slope $\sum_i s_{it}(n)\,\beta_i(n)\,(1 - w_{it}(n))$, for Treasuries and agency MBS from 1985 to 2025, and Table 5 reports the 2025Q1 slopes and impacts.

The Treasury slope is thin and has been thin since the turn of the century. In 2025Q1 it is 0.130 (standard error 0.049, $t = 2.7$), so removing 1 percent of the private float raises the 10-year yield by about 8 basis points, with a Fieller band of $[4.4, 28.9]$ that is finite at both ends. The composition explains the number. Foreign investors hold 40 percent of the float, and their slope is not distinguishable from zero. Banks, pension funds, and insurers together hold 16 percent and contribute 6 percent of the slope. The household residual holds 15 percent and, with its slope of 0.73, supplies 82 percent of it; the mutual funds hold 14 percent and supply a further 22 percent; the arbitrageurs subtract 13 percent and the remaining sectors contribute the last 3. The market's marginal buyers are two sectors holding less than a third of the float between them. Over time the slope tracks that sector's presence: it averaged 0.12 from 1985 to 1999 and fell to 0.042 between 2005 and 2008, reaching its low of 0.022 in 2005Q1, as foreign official reserve managers accumulated Treasuries and the residual emptied. It averaged 0.07 over the QE decade, touching 0.04 in 2014, and has averaged 0.12 since the float shifted back toward the residual, and the hedge funds inside it, after 2022. In the years when the slope is near zero the static system has almost no marginal buyer, and the yield effect of a given supply shift is correspondingly large; Section 5 shows what that means for the 2012–2014 and 2017–2019 episodes.

The agency MBS aggregate demand slope is four times larger and, unlike the Treasury slope, it is estimated with precision. In 2025Q1 it is 0.512 (standard error 0.140, $t = 3.7$), so removing 1 percent of the MBS float widens the option-adjusted spread by 2.0 basis points, with a Fieller band of $[1.3, 4.2]$ that is finite at both ends. The precision comes from the spread specification: under the yield specification the same calculation gives 0.36 with an

infinite upper bound. Every sector type contributes, because every slope is positive: banks hold 33 percent of the private float, mutual funds 17 percent, foreign investors 14 percent, and the household residual 11 percent, and none of them subtracts. The contrast with the Treasury market is the paper's central asymmetry in estimated form. Treasury absorption rests on two sectors holding less than a third of the float, and on one of them for four-fifths of the slope; agency MBS absorption is spread across all eight sector types. Appendix D.9 asks the complementary flow question, how much of the realized variation in each price supply accounts for, and finds the same contrast: supply explains about a sixth of the variance of the MBS spread and an eighth of the Treasury yield's.

The two markets are also linked, and the direction matters for what follows. Because the MBS block prices the spread, the level of the MBS yield is inherited from the Treasury market by construction: a supply shift in Treasuries passes into agency MBS yields one for one and the agency MBS market determines only the wedge on top. The two markets interact only through the logit shares, and that channel is small: in the $1 trillion scenario of Section 5, the $270 billion of MBS runoff alone raises the Treasury yield by 0.7 basis points, and the $730 billion of Treasury runoff alone moves the spread by -0.4. The well-identified part of the MBS result is therefore the spread, and the level component carries the Treasury market's wider uncertainty with it.

The system also implies the full set of own- and cross-price elasticities, and Panel A of Figure 3 plots them. Each of its two panels fixes a one-percent rise in one market's price and traces demand in both. The own elasticities are sizable and move with the price level: Treasury demand responds by 0.45 percent per percent on average and by 0.53 percent at the 2025Q1 composition, falling through the low-yield decade and recovering after 2022, because the elasticity carries the yield level that the aggregate demand slope does not. The agency MBS entries are elasticities with respect to the spread, which averages 0.56 percentage points against a Treasury yield that averages 4.8 percent, so their smaller size is the price level and not a weaker response; the underlying semi-elasticity is four times the Treasury one. The

cross elasticities sit near zero throughout, an order of magnitude below the own elasticities, because each scales with the share of the shocked asset in the portfolio and no bond is a large share of any sector's balance sheet. Substitution on the demand side therefore runs between bonds and the outside asset, not between the two bonds. The spillover the counterfactual reports is the general-equilibrium counterpart, in which each market re-clears, not direct substitution. Appendix Figure A.2 shows the same object by sector type and Appendix Table A.3 reports the matrix.

Table 5: **Aggregate Demand Slopes and Price Impact, 2025Q1**

Asset	Slope $\sum_i s_i\beta_i(n)(1-w_i(n))$	t	Impact per 1% of float (bp)	95% Fieller band
Treasury	+0.130 (0.049)	2.7	7.7	[4.4, 28.9]
Agency MBS	+0.512 (0.140)	3.7	2.0	[1.3, 4.2]

Notes: This table shows the aggregate demand slope of Equation (12) at the 2025Q1 composition of holdings, its delta-method standard error (sector-type slopes independent; the two cells of Section 3.7 contribute none), and the implied yield impact of removing 1 percent of the private float, 1/slope, in basis points, with the Fieller band from the slope's 95 percent interval (unbounded above when that interval includes zero).

5 Counterfactual Simulation

5.1 Setup

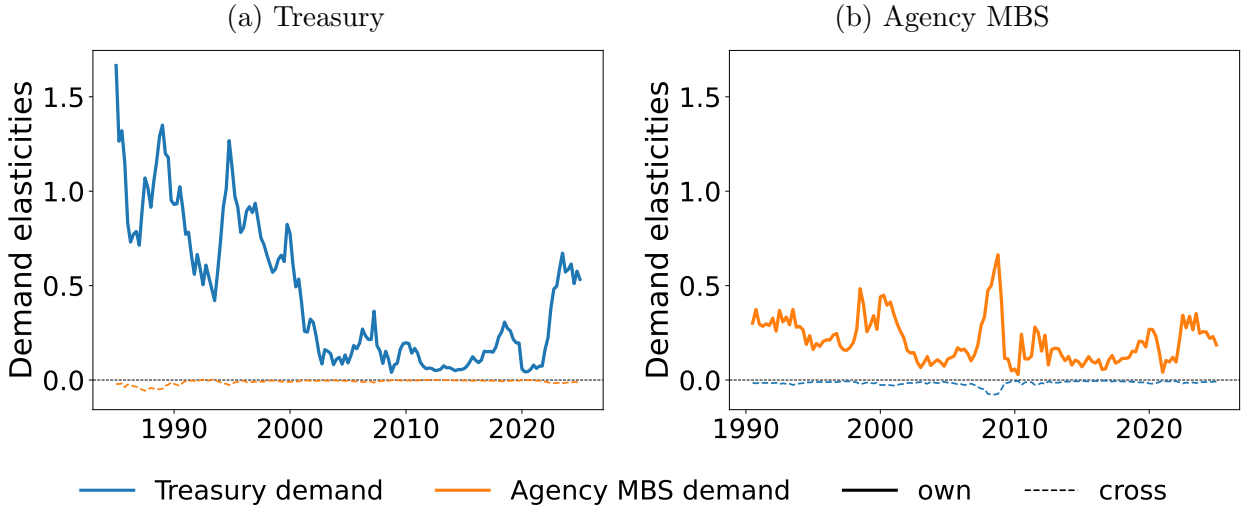
The equilibrium asset yields are jointly determined by the estimated demand system in (8) and the market-clearing condition in (13). To close the asset pricing model, the market for each asset n clears in each period t:

$$\sum_{i=1}^{I} w_{it}^{*}(\boldsymbol{y}_t^{*},\, \boldsymbol{x}_t,\, \boldsymbol{\beta}_i,\, \boldsymbol{\varepsilon}_{it}) \cdot A_{it} = S_t(n;\, \boldsymbol{y}_t^{*}), \tag{17}$$

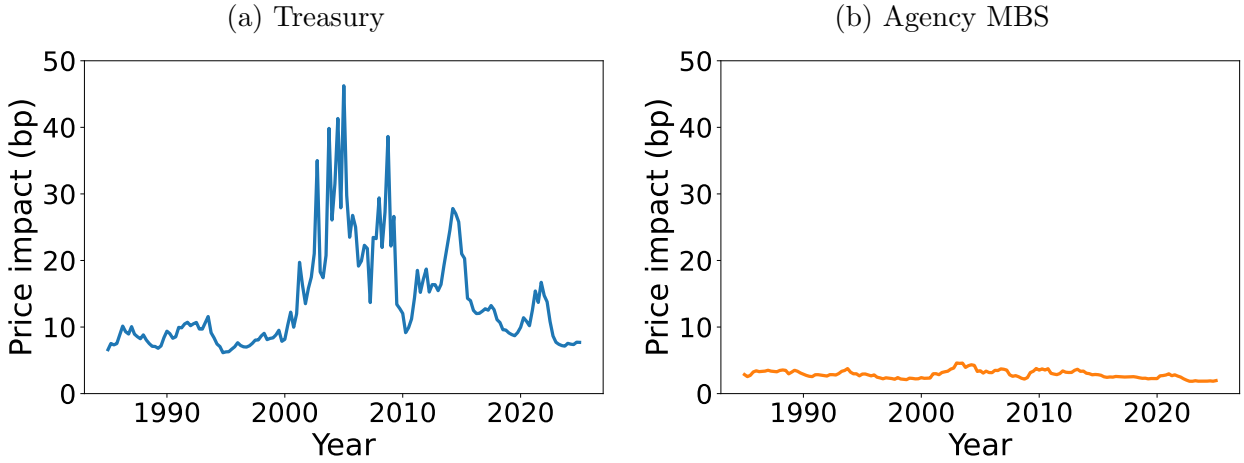
where $\boldsymbol{y}_t^{*}$ is the equilibrium yield vector, $\boldsymbol{x}_t$ is the asset-characteristics vector, $\boldsymbol{\beta}_i$ collects investor i's asset-specific demand slopes, $\boldsymbol{\varepsilon}_{it}$ are the latent demand shocks, and $S_t(n;\, \boldsymbol{y}_t^{*})$ is the supply schedule of asset n, $S_t(n;\, \boldsymbol{y}) = f_n\big(\boldsymbol{y};\, \bar{S}_t(n)\big)$, which lets net issuance respond

Figure 3: **Bond Demand Elasticity and Implied Price Impact**

Panel A: **Market Demand Elasticities**



Panel B: **Implied Price Impact of 1% of Private Float**



Notes: This figure shows, in Panel A, the holdings-weighted market elasticities $\eta_{it}(n,n) = \beta_i(n)(1-w_{it}(n))\, y_t(n)$ (own, solid) and $\eta_{it}(m,n) = -\beta_i(m) w_{it}(m)\, y_t(m)$ (cross, dashed) for a one-percent rise in each market's own $y_t(n)$, the 10-year yield for Treasuries and the option-adjusted spread for agency MBS (from 1990Q3), and in Panel B the yield impact of removing 1 percent of the private float, $1/\sum_i s_{it}(n)\,\beta_i(n)\,(1-w_{it}(n))$ in basis points (Equation 12), point estimates only (Table 5 gives the 2025Q1 band). Both panels use the one-year slopes of Table 4 with the cell models of Section 3.7 and floor the household Treasury residual at zero (Appendix D.2).

to the equilibrium yield; $\bar{S}_t(n)$, the private float at baseline yields, is the schedule's shifter, and at baseline yields supply equals the observed float. Section 5.2 estimates the schedule's reduced form. The equilibrium yield vector $\boldsymbol{y}_t^*$ is the implicit function

$$\boldsymbol{y}_t^* = \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t\big), \tag{18}$$

where $\boldsymbol{g}(\cdot)$ is the implicitly defined equilibrium mapping, which clears every market against the supply schedule, $\bar{\boldsymbol{S}}_t$ is the vector of baseline floats, and $\boldsymbol{A}_t$ is the vector of investor assets under management.

The system is nearly block-recursive in the two markets the Fed trades. Treasury demand responds to the 10-year yield and agency MBS demand to the option-adjusted spread (Section 3.6), so the Treasury market clears the level, the agency MBS market clears the spread, and the mortgage-relevant yield is the sum of the two,

$$\Delta y_t^*(\text{MBS}) = \underbrace{\Delta y_t^*(\text{Treasury})}_{\text{level, from the Treasury market}} + \underbrace{\Delta s_t^*}_{\text{spread, from the agency MBS market}}. \tag{19}$$

Supply is the one place the level re-enters the MBS block: origination responds to the mortgage rate, not to the spread, so the supply schedule of Equation (22) is evaluated at the reconstructed yield. Equation (19) is the decomposition the counterfactual reports throughout; a specification that prices MBS demand on the yield level cannot produce it, because its MBS market clears a price that already contains the Treasury level.

I impose clearing in those two markets.

I evaluate the impact of counterfactual monetary policy by comparing baseline and counterfactual equilibrium yields. Changes in the Fed's asset holdings translate into changes in the baseline private private float $\bar{\boldsymbol{S}}_t$, and I measure them as the Fed's transactions over the episode, the sum of its quarterly net purchases in the Z.1 flow tables, rather than as the change in the market value of its portfolio, which in 2022 mixes runoff with the revaluation of the securities it held. This one-for-one translation holds the fiscal path fixed: debt the

Fed does not roll over is refinanced to the public at auction, and agency MBS paydowns are re-securitized, so runoff adds to the float exactly as outright sales would; any yield-induced change in gross issuance is what the supply schedule captures. Denoting the post-purchase baseline float by $\bar{\boldsymbol{S}}_t^{\mathrm{CF}}$, the yield effect is

$$\Delta \boldsymbol{y}_t = \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t^{\mathrm{CF}}\big) - \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t\big). \tag{20}$$

The logit system is nonlinear. If every slope $\beta_i(n)$ is positive, excess demand is monotone in the yield and the equilibrium is unique. The estimates do not satisfy that condition, since the dealer cell and the impact-quarter residual carry negative slopes, so I rely on the weaker requirement that the aggregate demand slope be positive along the path the solver follows; it is at every date for the long-run slopes, and it fails in the impact quarter of the agency MBS market. With negative slopes the system can have several roots, and I solve it by continuation, scaling the supply shift from zero to its full size in small steps and following the equilibrium that is continuously connected to the observed one. Where that path breaks, the shock is larger than the market's absorbers can take at that date's composition of holdings, and I report the first-order impact of Equation (12) instead, flagged as such.

5.2 Supply Response

The counterfactual lets bond supply respond to yields: when the counterfactual yield rises, issuers issue less (Fang and Xiao, 2025), attenuating the yield effect. This supply side is deliberately reduced-form, a first-order net-issuance elasticity rather than a structural model of issuer behavior, since the demand system is the paper's structural focus. I estimate these elasticities from the same Flow-of-Funds data as the demand system. The identification mirrors the demand side. Estimating a demand slope requires supply-side variation in yields; estimating the supply (issuance) response requires *demand*-side variation, so the supply

regressions use demand shifters, described below, rather than the demand system's own instruments. The specification parallels the local projections of the demand system, with the response read at the same one-year horizon $h = 4$ as the demand slope $\beta^{(4)}$: the cumulative net issuance, measured from transactions as a percent of outstanding, through one year after a one-quarter yield change,

$$\sum_{s=t}^{t+4} \frac{F_s(n)}{S_{s-1}(n)} \times 100 = \kappa_n^{(4)} \, \Delta y_t(n) + \text{quarter fixed effects} + \tau \, t + u_t(n), \tag{21}$$

with $\Delta y_t(n)$ instrumented, quarter fixed effects absorbing the Treasury refunding cycle. The implied supply elasticity is $\psi_n = -\kappa_n^{(4)}$, the percent of outstanding withdrawn per percentage-point yield rise per year. The counterfactual's supply schedule is its first-order form,

$$S_t(n;\, \boldsymbol{y}) = \bar{S}_t(n) \left[1 - \psi_n \, \Delta y_t(n)/100\right], \tag{22}$$

so relative to the baseline float $\bar{S}_t(n)$, issuers withdraw ψ_n percent of outstanding per percentage point that the counterfactual yield rises above its baseline value.

A supply curve has to be traced out by shifts in demand, just as a demand curve has to be traced out by shifts in supply. The coupon heteroskedasticity instrument identifies the demand side and so cannot identify supply: an instrument excluded from demand and correlated with the yield through supply cannot also be excluded from supply and correlated with the yield through demand. Used in the supply regression it returns a positive coefficient, +5.6 with a standard error of 5.8. I instrument instead with Japanese Ministry of Finance foreign-exchange intervention. When the Ministry sells yen and buys dollars it places the proceeds in Treasuries, so the shock originates in Japanese exchange-rate policy rather than in any decision of the U.S. Treasury, and its first stage on the 10-year yield in the supply regression is 19.4 (16.4 in the demand block, Appendix Table A.8). The sign flips: Treasury net issuance falls by 3.2 percent of outstanding per percentage point of yield per year, with a standard error of 6.1. The Fed's purchase-path revision is a demand shifter too, and I do not

use it: it returns -27, more than a quarter of the outstanding stock per percentage point, it survives a control for federal net borrowing, and it is identified off 2009 and 2020, when the purchase announcements and the fiscal expansions arrived together. Agency MBS issuance responds by -1.1 percent of outstanding per percentage point per year (standard error 3.4), identified by the high-frequency MBS-price surprise around FOMC announcements of Drukker (2025). Fang and Xiao (2025) estimate the same aggregate object on the same kind of data and report 0.96 for Treasuries and 2.24 for agency MBS, but against the term spread and the MBS spread rather than the yields, so their numbers are a reference point and not a like-for-like benchmark; I do not carry them in the table. Measured from the change in market-value outstanding, every issuance elasticity is larger and more precise, because the market-value change contains the revaluation $-D\,\Delta y$ and so responds to yields by construction. The Treasury value is the one that matters for the headline numbers. At $\psi = 3.2$ the supply side absorbs a meaningful part of a Fed purchase. It absorbs the same part whatever the length of the program. One could scale ψ by the horizon, on the view that issuance keeps responding for as long as the yield stays away from baseline, but the estimated path says the response is a stock adjustment rather than a flow. The Treasury $\kappa^{(h)}$ peaks at one year, -3.2, and is back above zero by two, $+1.8$, where linear accumulation would have predicted -6.5. Each episode therefore receives $\kappa^{(4)}$ once. The choice matters for the comparison of programs of similar size: QE2 and QE3 bought \$808 billion and \$817 billion of Treasuries, and under horizon scaling the ten-quarter program looked a third the size of the three-quarter one.

5.3 Assessment of Historical QE/QT Episodes

I apply the structural counterfactual to the four QE programs (QE1–QE4) and the two QT episodes (QT1, 2017–2019; QT2, 2022–2025). For each episode I hold investor characteristics, long-run demand slopes, and AUM at their observed values, let bond supply respond to yields through the net-issuance elasticities of Section 5.2, and ask how yields would differ had the

Table 6: **Net-Issuance Elasticities Estimated from the Flow of Funds**

	One-year response $\kappa^{(4)}$		First-stage	Implied ψ
Asset	Coefficient	(s.e.)	F	(%/pp/yr)
Treasury	−3.24	(6.09)	19.4	3.24
Agency MBS	−1.12	(3.42)	2.5	1.12

Notes: This table shows instrumental-variables local projections of cumulative net issuance (transactions, percent of lagged outstanding) over the year after a one-quarter yield change, per asset, 1985Q1–2025Q1, with Newey–West standard errors in parentheses and the crisis indicator as a control. The Treasury yield is instrumented by Japanese foreign-exchange intervention (first stage 19.4); the agency MBS yield, on which origination responds, by the Drukker (2025) monetary surprise with the transaction-based issuance interaction (first stage 2.5, so the agency MBS elasticity is weakly identified). The implied ψ is the supply elasticity used in the counterfactual, set to zero where the coefficient is positive. *, **, ***: $|t|$ above 1.64, 1.96, 2.58.

Fed not transacted. Table 7 reports the long-run effects, at the one-year slopes.

Four findings stand out. First, the Treasury effects are large, and they scale with the shock. QE1's $300 billion of Treasury purchases compressed the 10-year yield by 54 basis points, QE2's $808 billion by 103, QE3's $817 billion by 112, and QE4's $3.5 trillion by 186; scaled by the share of the private float removed (final column), the programs run from 8 to 12 basis points per 1 percent of float, net of the issuance the supply side returns to the market. They are large because the Treasury market's absorber is thin: the household residual and the mutual funds, less than a third of the float between them, are the only sectors that take the other side, and the foreign, bank, and pension holdings that dominate the float do not move. QE2 and QE3 bought almost the same amount of Treasuries, $808 billion against $817 billion, and move the yield by 103 and 112 basis points. QE2 took a larger share of a smaller float, 12.2 percent against 9.2, but QE3 ran when the Treasury aggregate demand slope was lower, so each percent of float moved yields further.

Second, the agency MBS yield moves with the Treasury yield, and what the MBS market itself determines is the spread. Equation (19) splits each entry, and the split is the finding. QE1, the Fed's only MBS-heavy program at $1.2 trillion against $300 billion of Treasuries, compressed the MBS yield by 108 basis points, of which 54 is the Treasury level and 54 is

spread compression: this is the one program whose composition was mortgage-specific, and it is the only one with a large spread effect. QE3 and QE4 bought both markets in rough proportion and moved the spread by 31 basis points each; their MBS yield effects, 143 and 217 basis points, are mostly the Treasury level passing through. QE2 is the mirror image of QE1: the Fed bought $808 billion of Treasuries while running off $207 billion of MBS, which *widened* the spread by 12 basis points, so agency MBS yields fell 90 basis points against the Treasury market's 103 and homeowners received seven-eighths of the pass-through. In this system, composition governs what QE does to the mortgage spread and size what it does to the level.

The QE1 estimate is also where the demand system now meets the event-study record. Its 108 basis points sit just above the range between Hancock and Passmore (2011) (−85) and Krishnamurthy and Vissing-Jorgensen (2011) (−107). The reconciliation is mechanical: an MBS demand curve written on the yield level makes the MBS market absorb the whole level shock by itself, and the elastic absorbers then deliver a small number; written on the spread, the level arrives from the Treasury market and only the wedge is absorbed locally.

Third, QE4 is the largest program because its shock was largest, not because demand was steeper. It removed a quarter of the private Treasury float, and scaled by that shock it moved yields by 7.7 basis points per 1 percent, the lowest of the QE programs. Two-thirds of its Treasury purchases came in the first half of 2020, when the Fed bought in weeks to restore market functioning (Vissing-Jorgensen, 2021); rebasing the counterfactual to the stock-QE phase that followed isolates a $1.2 to $1.5 trillion program that moved Treasury yields by 82 to 96 basis points, about 10 basis points per 1 percent, because the residual's share of the float was low in 2021.

Fourth, the QT episodes reverse the sign. QT1's $373 billion of Treasury runoff and $318 billion of MBS runoff raised the two yields by 23 and 31 basis points, 8 per 1 percent of float in each market, with 9 basis points of that coming from the spread. QT2's $1.6 trillion of Treasury runoff raised the 10-year yield by 65 basis points and the agency MBS

yield by 76, of which 10 is spread. The treatment of the two cells in Section 3.7 matters for this episode: with money funds pooled into the mutual-fund type and the dealer cell carrying a free elasticity, the Treasury aggregate demand slope is small enough that no equilibrium connected to the observed one exists at QT2's size. With the two cells modelled, the nonlinear system clears at every episode. The message is the one Section 4.4 anticipated: at the current composition of the float the Treasury market's aggregate demand slope is small enough that a shock of QT2's size is at the edge of what a static demand system can price.

Table 7: **Yield Effects of Historical QE/QT Episodes**

(a) Quantitative easing

Episode	Asset	Purchases ($B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
QE1	Treasury	300	−54.4	—	10.4
2008Q4–2010Q1	Agency MBS	1,223	−108.4	−54.0	5.5
QE2	Treasury	808	−102.7	—	8.4
2010Q4–2011Q2	Agency MBS	−207	−90.2	12.4	28.1
QE3	Treasury	817	−112.4	—	12.2
2012Q3–2014Q4	Agency MBS	865	−143.0	−30.6	10.4
QE4	Treasury	3,516	−186.0	—	7.7
2020Q1–2021Q4	Agency MBS	1,238	−216.8	−30.8	14.5

(b) Quantitative tightening

Episode	Asset	Runoff ($B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
QT1	Treasury	−373	22.7	—	7.8
2017Q4–2019Q3	Agency MBS	−318	31.2	8.5	8.1
QT2	Treasury	−1,622	65.4	—	7.7
2022Q2–2025Q1	Agency MBS	−549	75.6	10.3	13.3

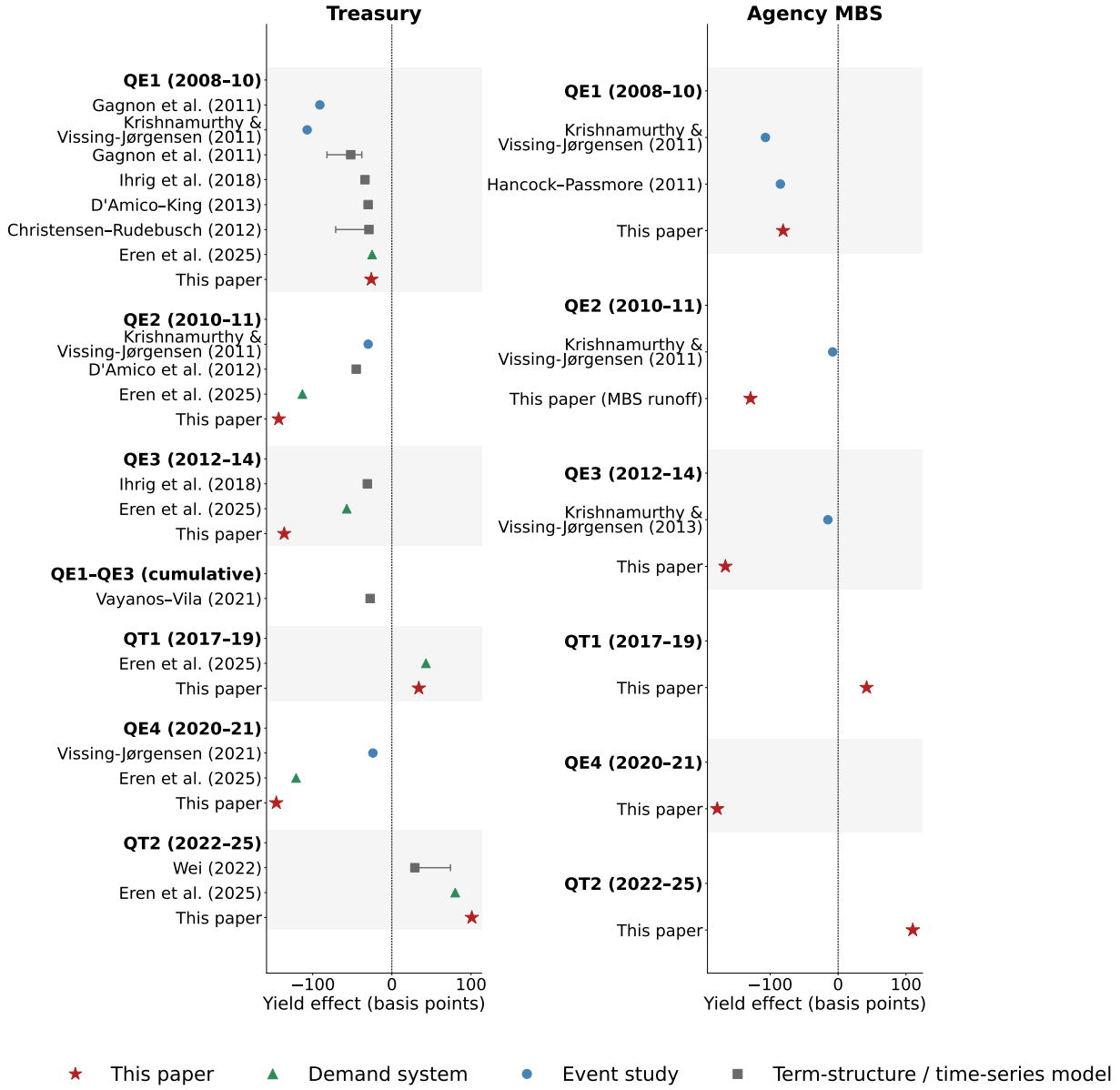
Notes: This table shows the yield effects, in basis points, of holding the Fed's portfolio at its pre-episode level, from the market-clearing counterfactual of Section 5: the float is shifted by the Fed's transactions over the episode and both markets are re-solved against the supply schedule of Equation (22), measured at the episode-end quarter. "Of which spread" is the spread term of Equation (19); the last column is $|\Delta Y|$ per 1 percent of the private float. The two cells of Section 3.7 follow their own models.

Per unit of supply removed, the Treasury estimates agree with the demand-system literature. Scaled by the shock they run 8 to 12 basis points per 1 percent of float, net of the issuance the supply side returns, against the 8 to 13 basis points per percentage point of the central bank's share that Eren et al. (2026) report and the 13 basis points per 1 percent shift in demand of Chaudhary et al. (2026). Figure 4 places the estimates episode by episode. The two systems are close on QE2 (103 here against their 113), and mine reads QE4 and QE3 larger (186 against 121, and 112 against 57). On the QT side mine are smaller, 23 against 43 for QT1 and 65 against 80 for QT2. Announcement-window event studies (Gagnon et al., 2011; Krishnamurthy and Vissing-Jorgensen, 2011) are larger for QE1 and smaller for the later, largely anticipated programs; they price only the surprise component of purchases, on impact, and include the signaling channel that the supply-only counterfactual excludes. QE1 enters here at 54 basis points, above the 25 of Eren et al. (2026), and between the calibrated preferred-habitat estimate of 25 to 30 for the cumulative QE1 to QE3 purchases (Vayanos and Vila, 2021) and the event-study range of 91 to 107. On the mortgage side the agreement is closer: QE1's 108 basis points lie at the top of the announcement-window range for agency MBS.

Three caveats bound the interpretation, and they do not all run in the same direction. The counterfactual holds latent demand and AUM fixed and treats the supply change as permanent, so the long-run estimates overstate the effect if investors expected the Fed to unwind (Jansen et al., 2024); the demand slopes are identified from local variation, while the episodes shift supply well outside that range; and the aggregate Treasury slope has a wide band, so the per-unit impact of 7.7 basis points at the 2025Q1 composition carries a 95 percent Fieller band of $[4.4, 28.9]$ (Table 5), finite at both ends but wide at the top. Weak-instrument bias runs the same way: it pulls the absorbers' slopes toward zero, which makes the yield effects, if anything, too large. The two components of the mortgage number are bounded differently. The spread component is the tightest object in the paper: 2.0 basis points per 1 percent of the MBS float, with a two-sided Fieller band of $[1.3, 4.2]$. The

level component is the Treasury number and inherits its band, so a reader who doubts the Treasury absorber should read the mortgage result as the spread effect plus whatever the Treasury market delivers.

Figure 4: **Historical QE/QT Yield Effects: This Paper versus Prior Estimates**



Notes: This figure shows the estimates of Table 7 (stars) against prior estimates for each episode, Treasuries (left) and agency MBS (right): circles are event studies, squares term-structure and time-series models, triangles demand systems; bars are ranges reported by the source. The Vayanos and Vila (2021) entry is a calibration of the cumulative QE1–QE3 purchases. Episode definitions differ slightly across studies.

5.4 Prospective QT Scenario Analysis

I next project the yield effects of a resumption of quantitative tightening. Fixing the 2025Q1 portfolio and running additional runoff at the QT2 Treasury:MBS ratio, I consider three scenarios: a mild $400 billion, a baseline $1 trillion, about a year and a half at the Fed's 2024 runoff pace of $172 billion per quarter, and an aggressive $1.6 trillion. Table 8 reports the results. A further $1 trillion raises 10-year Treasury yields by 22 basis points and agency MBS yields by 27, of which 5 basis points is spread widening and the rest is the Treasury level passing through; the mild scenario raises them by 9 and 11, the aggressive by 35 and 42. A back-of-the-envelope calculation conveys the real stakes. Applied to the roughly $27 trillion of marketable federal debt as it reprices, the baseline Treasury rise adds about $60 billion a year to federal interest costs, a recurring fiscal cost to weigh against the balance-sheet reduction. On the household side, agency MBS yields pass through nearly one for one to primary mortgage rates, so the baseline adds about $72 a month to the payment on a newly originated $400,000 thirty-year fixed-rate mortgage.[4] The marginal Treasury impact is nearly constant, about 2 basis points per $100 billion of additional runoff. The mortgage cost of quantitative tightening is therefore mostly a Treasury-market phenomenon: a runoff split 73:27, close to QT2's 75:25, leaves the mortgage spread almost unchanged, and the homeowner pays for the level.

The Treasury number is conditional on the marginal buyer, and I report it as a band rather than a point. Two cells carry the Treasury aggregate demand slope, and neither is pinned down. The household residual's slope of 0.73 supplies four-fifths of the aggregate at the 2025Q1 composition, and it is identified on the pre-2010 sample: in the 2010–2025 window the sector's slope is -0.84 with a standard error of 0.82 (Section 4). The foreign sector carries a larger weight still in the aggregate demand slope, because it holds 40 percent of the float, and its slope is zero under the paper's instruments but 0.33 under the Fed

[4] Fiscal: $27 trillion $\times$ 22.4 basis points $\approx$ $60 billion per year once the stock has repriced. Mortgage: the monthly payment on a loan of size P at annual rate r is the standard annuity $M(r) = P\,(r/12)\,/\,[1 - (1 + r/12)^{-360}]$; with $P = \$400{,}000$ and $r = 6.5\%$, $M(r + 27\,\text{bp}) - M(r) \approx \$72$ per month.

Table 8: **Yield Effects of Prospective Quantitative Tightening**

(a) Baseline

Scenario	Asset	Runoff ($B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
Mild ($400B)	Treasury	−292	9.4	—	6.2
	Agency MBS	−108	11.3	2.0	10.1
Baseline ($1T)	Treasury	−730	22.4	—	5.9
	Agency MBS	−270	27.3	4.8	9.7
Aggressive ($1.6T)	Treasury	−1,168	34.5	—	5.7
	Agency MBS	−432	42.2	7.6	9.4

(b) Sensitivity to Treasury slopes

Scenario	Asset	Runoff ($B)	ΔY (bp)	of which spread (bp)	ΔY per 1% of float (bp)
Expansion slope	Treasury	−730	34.8	—	9.2
$\beta_{\text{HR}}^{\text{Treasury}}$: baseline → 0.34	Agency MBS	−270	39.4	4.6	14.0
Recession slope	Treasury	−730	6.2	—	1.6
$\beta_{\text{HR}}^{\text{Treasury}}$: baseline → 3.58	Agency MBS	−270	11.3	5.2	4.0
Lower AR bound	Treasury	−730	44.8	—	11.8
$\beta_{\text{HR}}^{\text{Treasury}}$: baseline → 0.18	Agency MBS	−270	49.2	4.4	17.5
Higher foreign slope	Treasury	−730	14.9	—	3.9
$\beta_{\text{Foreign}}^{\text{Treasury}}$: baseline → 0.33	Agency MBS	−270	20.2	5.3	7.2
Two-year slopes	Treasury	−730	13.6	—	3.6
all slopes: $\beta^{(4)} \to \beta^{(8)}$	Agency MBS	−270	20.0	6.4	7.1

Notes: This table shows projected yield effects, in basis points, of runoff beyond the 2025Q1 balance sheet, split 73:27 as in QT2, from the nonlinear clearing system of Section 5 with the supply elasticities of Table 6. Panel (a) uses the baseline slopes. Panel (b) re-solves the $1 trillion row under the slope named in each row label, every other slope at baseline: the household residual's expansion and recession slopes (Appendix D.7) and the lower edge of its Anderson–Rubin set, a foreign slope of 0.33 (Appendix Table A.8), and the two-year slopes $\beta^{(8)}$.

purchase-path revision alone (Appendix Table A.8). Panel (b) of Table 8 re-solves the baseline scenario across both.

The household dimension is the one that can make the effect large, and the business cycle is what moves it. The residual sector's slope differs across states of the cycle more than any other cell's: 0.34 in expansions against 3.58 in NBER recessions, a difference significant at the 10 percent level ($p = 0.064$) (Appendix D.7). A runoff carried out in an expansion therefore raises the 10-year yield by 35 basis points and the same runoff in a recession by 6, against 22 at the pooled slope. The statistical band runs the same way: at the lower edge of the slope's Anderson–Rubin set, 0.18, the effect is 45 basis points. That edge has an economic reading, not only a statistical one. The residual sector is now mostly the hedge funds running the cash–futures basis trade, which hold 122 percent of its coupon Treasuries on average since 2013 (Section 4.2), while the slope applied to it is identified on the pre-2010 sample, when the same residual was households and nonprofits. A low household slope is the case in which the basis trade absorbs less readily than the buy-and-hold investor it replaced. The foreign dimension runs the other way: with foreign investors at 0.33 the same runoff raises Treasury yields by 15 basis points rather than 22. So does the horizon: several absorbers keep building through the second year (Section 4.3), and with every slope read at two years the runoff raises Treasury yields by 14 basis points and agency MBS yields by 20. The honest range for a $1 trillion runoff is therefore 6 to 45 basis points on the 10-year yield, every value of it a genuine equilibrium, and which end applies turns on two sector slopes that four decades of quarterly data do not separate. The Fieller band on the first-order impact tells the same story from the aggregate side: $1 trillion is 3.8 percent of the Treasury float, and before any supply response the impact per 1 percent is 7.7 basis points with a 95 percent interval of $[4.4, 28.9]$.

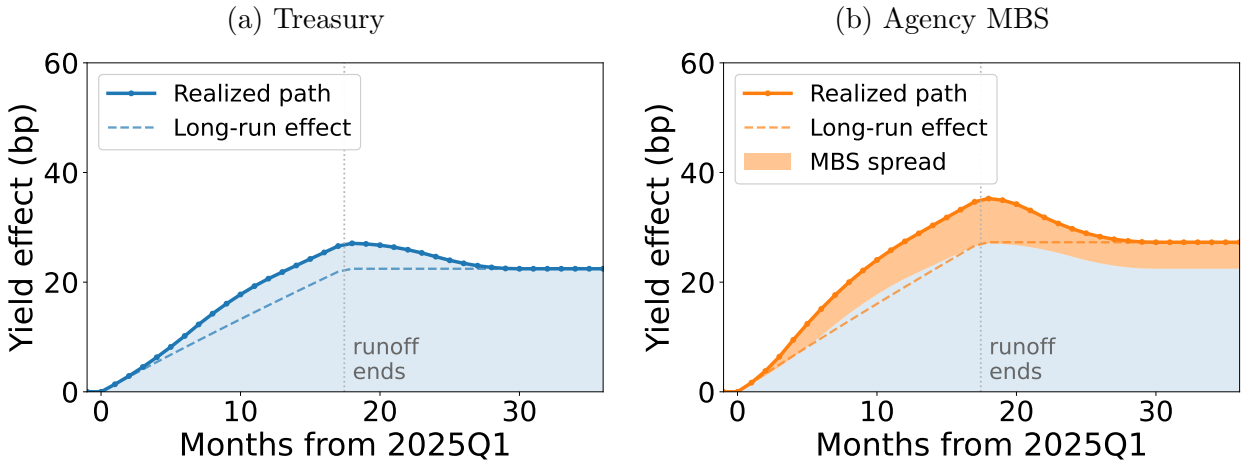
The mortgage number splits along the same seam, and the split is what makes it readable. The spread component barely moves across the band, 4 to 6 basis points in every row of Panel (b), because the MBS absorbers are identified in both windows and none of them is

the residual. Its own interval is two-sided, $[1.3, 4.2]$ basis points per 1 percent of the MBS float. The level component is the Treasury number, so the agency MBS yield rises by 27 basis points at the pooled slope, 11 at the recession slope, 39 at the expansion slope, and 49 at the lower Anderson–Rubin edge. The honest summary is that the mortgage-specific piece of quantitative tightening is small and tightly estimated, while the mortgage cost of quantitative tightening is large and uncertain, because it is the Treasury market's number in disguise, and it rests on the assumption that the sector which absorbed Treasuries before 2010, the household residual and the hedge funds it now contains, will absorb them again.

Figure 5 projects the resulting yield path, combining the runoff pace with the estimated rebalancing dynamics. Each quarter's incremental runoff first lands on the impact demand curve: re-solving the \$1 trillion scenario horizon by horizon as the absorbers travel from $\beta^{(0)}$ to $\beta^{(4)}$, the impact quarter has no equilibrium at all, because the agency MBS aggregate slope is negative on impact and the Treasury slope is near zero, and the path peaks one to two quarters after the shock at about 1.6 times the long-run Treasury effect and 1.9 times the long-run mortgage effect, then decays to the long run within a year as slow capital arrives. Cumulating these vintages at the actual pace, Treasury yields rise by about one and a half basis points per month while the runoff proceeds, peak at 27 basis points when it ends after 18 months, and are still 25 two years out, on their way to the long-run 22; the mortgage path peaks at 35 and is 30 two years out, against a long-run 27. The projection says most of the yield effect of a resumed QT arrives *during* the runoff, roughly in proportion to the stock withdrawn, and that the gradual pace limits the transient: at any time only the most recent increments are mid-adjustment, so the path peaks only a few basis points above its long-run component. The impact overshoot and its decay are the price-impact counterpart of the horizon duality van der Beck (2026) documents for equities, and the adjustment dynamics of Greenwood et al. (2018), in which a supply shock is absorbed first by local specialists and only gradually by cross-market capital; Jiang and Sun (2024) derive the same overshooting for QT when long-term investors adjust slowly, and their prescription that runoff be slow

and steady is what the projection shows quantitatively.

Figure 5: **Projected Yield Path of a Resumed QT**



Notes: This figure shows the projected yield effects of $1 trillion of further runoff from the 2025Q1 portfolio at $172 billion per quarter (drawn monthly), split at the QT2 ratio. Each month's runoff enters at its impact effect and converges to its long-run effect as the absorbers move from $\beta^{(0)}$ to $\beta^{(4)}$; the dashed line is the accumulated long-run effect with the supply response of Section 5.2. The agency MBS panel is the yield, Treasury level plus spread: the orange band is the spread component and the blue area the Treasury level. The dotted line marks the end of the runoff, and the path converges to the long-run effect by month 30.

6 Conclusion

The balance sheet will be run down again, and the framework here is built for the questions a resumed runoff will raise: who takes the other side, how far yields rise, and how the pace shapes the path. Its answers rest on measuring demand from what investors actually trade. Measured that way, demand is inelastic for most investors, the few that absorb supply build their positions slowly, and the main Treasury absorber is the household residual, whose slope is identified mostly before 2010. Across the programs from QE1 to QT2, Treasury yields moved by 8 to 12 basis points per 1 percent of the float, and the Treasury market set the level of agency MBS yields while the MBS market set only the spread. Three caveats bound the answers. The Financial Accounts report no within-class maturity detail, so the system prices substitution across asset classes rather than along the curve; the supply side is

deliberately reduced-form; and the demand slopes are identified from local variation, while the largest programs shift supply well outside it. Within those bounds, a further $1 trillion of runoff from the 2025Q1 balance sheet widens the mortgage spread by a small and precisely estimated 5 basis points and raises Treasury yields by about 22. The Treasury number roughly doubles if the household residual absorbs Treasuries less readily than it once did, and it falls by two-fifths if the absorbers are given two years rather than one to rebalance; the gradual pace of the runoff keeps the transient above the long-run effect small.

References

Viral V. Acharya, Rahul S. Chauhan, Raghuram G. Rajan, and Sascha Steffen. Liquidity dependence and the waxing and waning of central bank balance sheets. NBER Working Paper 31050, National Bureau of Economic Research, 2024.

Tobias Adrian and Hyun Song Shin. Liquidity and leverage. *Journal of Financial Intermediation*, 19(3):418–437, 2010.

Heitor Almeida, Murillo Campello, Bruno Laranjeira, and Scott Weisbenner. Corporate debt maturity and the real effects of the 2007 credit crisis. *Critical Finance Review*, 1(1):3–58, 2012.

Sriya Anbil, Alyssa Anderson, Ethan Cohen, and Romina Ruprecht. Beyond reserves: The federal reserve's balance sheet and the repo market. Working paper, Federal Reserve Board. Previously circulated as "Stop Believing in Reserves", 2026.

Alyssa G. Anderson, Alessandro Barbarino, Anthony M. Diercks, and Stephen Miran. A user's guide to reducing the federal reserve's balance sheet. Finance and Economics Discussion Series 2026-019, Board of Governors of the Federal Reserve System, 2026.

Theodore W Anderson and Herman Rubin. Estimation of the parameters of a single equation in a complete system of stochastic equations. *The Annals of mathematical statistics*, 20 (1):46–63, 1949.

Isaiah Andrews, James H Stock, and Liyang Sun. Weak instruments in instrumental variables regression: Theory and practice. *Annual Review of Economics*, 11(1):727–753, 2019.

Steven Berry, James Levinsohn, and Ariel Pakes. A demand system for differentiated products. *Econometrica*, 63(4):841–890, 1995.

Carol C. Bertaut and Ruth Judson. Estimating U.S. cross-border securities positions: New

data and new methods. International Finance Discussion Papers 1113, Board of Governors of the Federal Reserve System, 2014.

Lorenzo Bretscher, Lukas Schmid, Ishita Sen, and Varun Sharma. Institutional corporate bond pricing. *The Review of Financial Studies*, 2025.

Indraneel Chakraborty, Itay Goldstein, and Andrew MacKinlay. Monetary stimulus and bank lending. *Journal of Financial Economics*, 136(1):189–218, 2020.

Manav Chaudhary, Zhiyu Fu, and Haonan Zhou. Anatomy of the treasury market: Who moves yields? Working paper, 2026.

Jens H. E. Christensen and Glenn D. Rudebusch. The response of interest rates to US and UK quantitative easing. *The Economic Journal*, 122(564):F385–F414, 2012.

Stefania D'Amico, William English, David López-Salido, and Edward Nelson. The Federal Reserve's large-scale asset purchase programmes: Rationale and effects. *The Economic Journal*, 122(564):F415–F446, 2012.

Marco Di Maggio, Amir Kermani, and Christopher J Palmer. How quantitative easing works: Evidence on the refinancing channel. *The Review of Economic Studies*, 87(3):1498–1528, 2020.

Leonel Diego Drukker. Do lower mortgage rates benefit first-time homebuyers? *Available at SSRN 5741124*, 2025.

Darrell Duffie. Presidential address: Asset price dynamics with slow-moving capital. *Journal of Finance*, 65(4):1237–1267, 2010.

Stefania D'Amico and Thomas B King. Flow and stock effects of large-scale treasury purchases: Evidence on the importance of local supply. *Journal of financial economics*, 108 (2):425–448, 2013.

Egemen Eren, Andreas Schrimpf, and Fan Dora Xia. The demand for government debt. *Management Science*, 2026.

Chuck Fang and Kairong Xiao. What do $40 trillion of portfolio holdings say about monetary policy transmission? *Available at SSRN 5025417*, 2025.

Xiang Fang, Bryan Hardy, and Karen K Lewis. Who holds sovereign debt and why it matters. *The Review of Financial Studies*, 2025.

William Fuchs, Satoshi Fukuda, and Daniel Neuhann. Demand-system asset pricing: Theoretical foundations. *Available at SSRN 4672473*, 2023.

William Fuchs, Satoshi Fukuda, and Daniel Neuhann. A trilemma for asset demand estimation. *Available at SSRN*, 2025.

Andreas Fuster and Paul Willen. $1.25 trillion is still real money: Some facts about the effects of the federal reserve's mortgage market investments. *FRB of Boston Public Policy Discussion Paper*, (10-4), 2010.

Joseph Gagnon, Matthew Raskin, Julie Remache, and Brian Sack. The financial market effects of the federal reserve's large-scale asset purchases. *International Journal of Central Banking*, 7(1):3–43, 2011.

Robin Greenwood, Samuel G Hanson, and Gordon Y Liao. Asset price dynamics in partially segmented markets. *The Review of Financial Studies*, 31(9):3307–3343, 2018.

Valentin Haddad and Tyler Muir. Market macrostructure: Institutions and asset prices. *Annual Review of Financial Economics*, 17:133–150, 2025. doi: 10.1146/annurev-financial-090524-120754.

Valentin Haddad, Zhiguo He, Paul Huebner, Péter Kondor, and Erik Loualiche. Causal inference for asset pricing. *Available at SSRN 5187305*, 2025.

Diana Hancock and Wayne Passmore. Did the federal reserve's mbs purchase program lower mortgage rates? *Journal of Monetary Economics*, 58(5):498–514, 2011.

Jarrad Harford, Sandy Klasa, and William F Maxwell. Refinancing risk and cash holdings. *The journal of finance*, 69(3):975–1012, 2014.

Jane Ihrig, Elizabeth Klee, Canlin Li, Min Wei, and Joe Kachovec. Expectations about the Federal Reserve's balance sheet and the term structure of interest rates. *International Journal of Central Banking*, 14(2):341–391, 2018.

Kristy AE Jansen, Wenhao Li, and Lukas Schmid. Granular treasury demand with arbitrageurs. Technical report, National Bureau of Economic Research, 2024.

Zhengyang Jiang and Jialu Sun. Quantitative tightening with slow-moving capital. Working paper, Northwestern University, 2024.

Òscar Jordà. Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182, 2005.

John Kandrac and Bernd Schlusche. Quantitative easing and bank risk taking: evidence from lending. *Journal of Money, Credit and Banking*, 53(4):635–676, 2021.

Narayana R Kocherlakota. Log-linear relative asset demand. Technical report, National Bureau of Economic Research, 2025.

Ralph SJ Koijen and Motohiro Yogo. A demand system approach to asset pricing. *Journal of Political Economy*, 127(4):1475–1515, 2019.

Ralph SJ Koijen and Motohiro Yogo. On the theory and econometrics of (demand system) asset pricing. *Available at SSRN 5274709*, 2025.

Ralph SJ Koijen, François Koulischer, Benoît Nguyen, and Motohiro Yogo. Inspecting the mechanism of quantitative easing in the euro area. *Journal of Financial Economics*, 140 (1):1–20, 2021.

Arvind Krishnamurthy and Annette Vissing-Jorgensen. The effects of quantitative easing on interest rates: Channels and implications for policy. *Brookings Papers on Economic Activity*, 43(2):215–287, 2011.

Arvind Krishnamurthy, Annette Vissing-Jorgensen, et al. The ins and outs of lsaps. In *Kansas City federal reserve symposium on global dimensions of unconventional monetary policy*, pages 57–111. Federal Reserve Bank of Kansas City Kansas City, 2013.

Robert Kurtzman, Stephan Luck, and Tom Zimmermann. Did qe lead banks to relax their lending standards? evidence from the federal reserve's lsaps. *Journal of Banking & Finance*, 138:105403, 2022.

Arthur Lewbel. Using heteroscedasticity to identify and estimate mismeasured and endogenous regressor models. *Journal of Business & Economic Statistics*, 30(1):67–80, 2012.

David López-Salido and Annette Vissing-Jorgensen. Reserve demand, interest rate control, and quantitative tightening. Working paper, Federal Reserve Board, 2025.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Junko Oguri and Cristoforo Pizzimenti. Ample reserves for whom? the role of foreign banks in u.s. monetary policy implementation. Working paper, Northwestern University, 2026.

Alexander Rodnyansky and Olivier M Darmouni. The effects of quantitative easing on bank lending behavior. *The Review of Financial Studies*, 30(11):3858–3887, 2017.

James H Stock and Mark W Watson. Identification and estimation of dynamic causal effects in macroeconomics using external instruments. *The Economic Journal*, 128(610):917–948, 2018.

Johannes Stroebel and John B Taylor. Estimated impact of the federal reserve's mortgage-backed securities purchase program. *International Journal of Central Banking*, 2012.

Eric T Swanson. Measuring the effects of federal reserve forward guidance and asset purchases on financial markets. *Journal of Monetary Economics*, 118:32–53, 2021.

Philippe van der Beck. Short versus long-run demand elasticities in asset pricing. *Journal of Financial Economics*, 184:104337, 2026.

Dimitri Vayanos and Jean-Luc Vila. A preferred-habitat model of the term structure of interest rates. *Econometrica*, 89(1):77–112, 2021.

Annette Vissing-Jorgensen. The Treasury market in spring 2020 and the response of the Federal Reserve. *Journal of Monetary Economics*, 124:19–47, 2021.

Kevin M. Warsh. Transcript of Chairman Warsh's press conference. Board of Governors of the Federal Reserve System, FOMC Press Conference, June 2026. URL https://www.federalreserve.gov/mediacenter/files/FOMCpresconf20260617.pdf. June 17, 2026; preliminary transcript.

Bin Wei. Quantifying "quantitative tightening" (QT): How many rate hikes is QT equivalent to? Policy Hub Paper 2022-8, Federal Reserve Bank of Atlanta, 2022.

A A BLP Interpretation of Sector-Level Asset Demand

In this appendix I give the logit demand of Section 3 a discrete-choice foundation in the spirit of BLP. Each sector i aggregates many institutions, but the data are sector-level portfolios. Treat each dollar of sector i's wealth as held by an investor h who places it in one asset. The indirect utility of investor h in sector i from holding asset n at time t is

$$u_{hit}(n) = \beta_i(n)\, y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n) + \ln \epsilon_{it}(n) + v_{hit}(n), \tag{23}$$

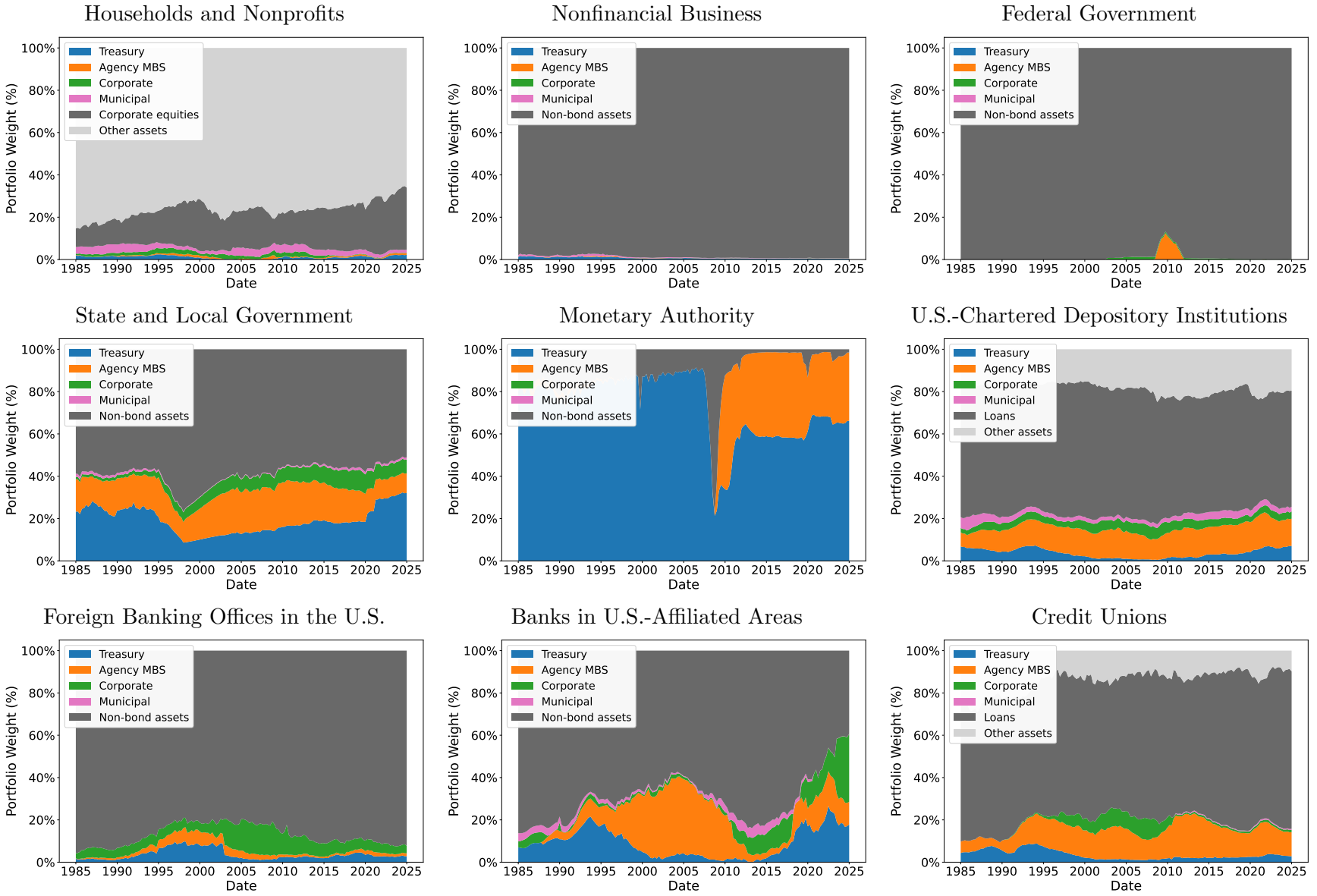
and the outside asset yields $u_{hit}(0) = v_{hit}(0)$. Here $\beta_i(n)$ and γ_i are preference parameters common to all investors in sector i; $\xi_{it}(n)$ is the sector–asset–time demand shifter of Section 3, parameterized there with macro controls; $\epsilon_{it}(n)$ is latent demand, common to the sector's investors; and $v_{hit}(n)$ are i.i.d. Type-I extreme-value shocks that generate idiosyncratic within-sector heterogeneity. Each sector–quarter is a market in the BLP sense: its investors face the same yields and characteristics and differ only through v. Integrating the extreme-value shocks out gives the fraction of the sector's wealth placed in asset n,

$$w_{it}(n) = \frac{\exp\{\beta_i(n)\, y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n)\}\, \epsilon_{it}(n)}{1 + \sum_{k=1}^{N} \exp\{\beta_i(k)\, y_t(k) + \gamma_i' x_t(k) + \xi_{it}(k)\}\, \epsilon_{it}(k)},$$

which is Equation (5) with the $\delta_{it}(n)$ of Equation (6); the shocks v do not appear in the share, and the latent demand $\epsilon_{it}(n)$ plays the role of the unobserved product characteristic in BLP. Random coefficients, standard in BLP, could in principle be identified from aggregate shares: a specification $\beta_{hi}(n) = \bar{\beta}_i(n) + \sigma_i(n)\, \nu_h$ would be traced out by the variation in yields, characteristics, and shares across quarters. Doing so requires rich variation in asset characteristics and valid instruments to identify the distribution separately from its mean, which security-level holdings provide (Fang and Xiao, 2025) and two asset classes observed at the sector level do not. I therefore restrict within-sector heterogeneity to the idiosyncratic choice shocks.

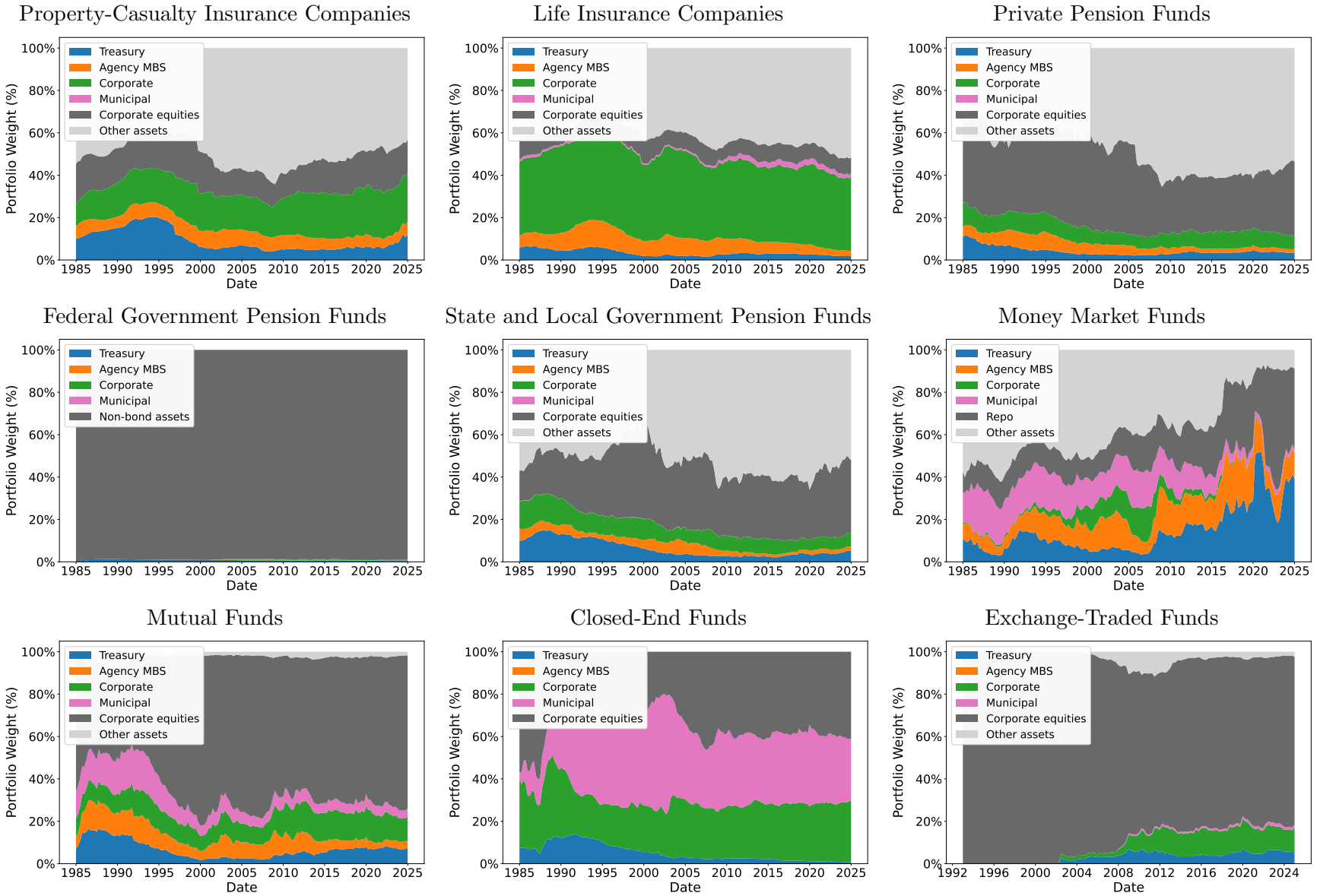
B Figures

Figure A.1: **Portfolio Shares by Sector**



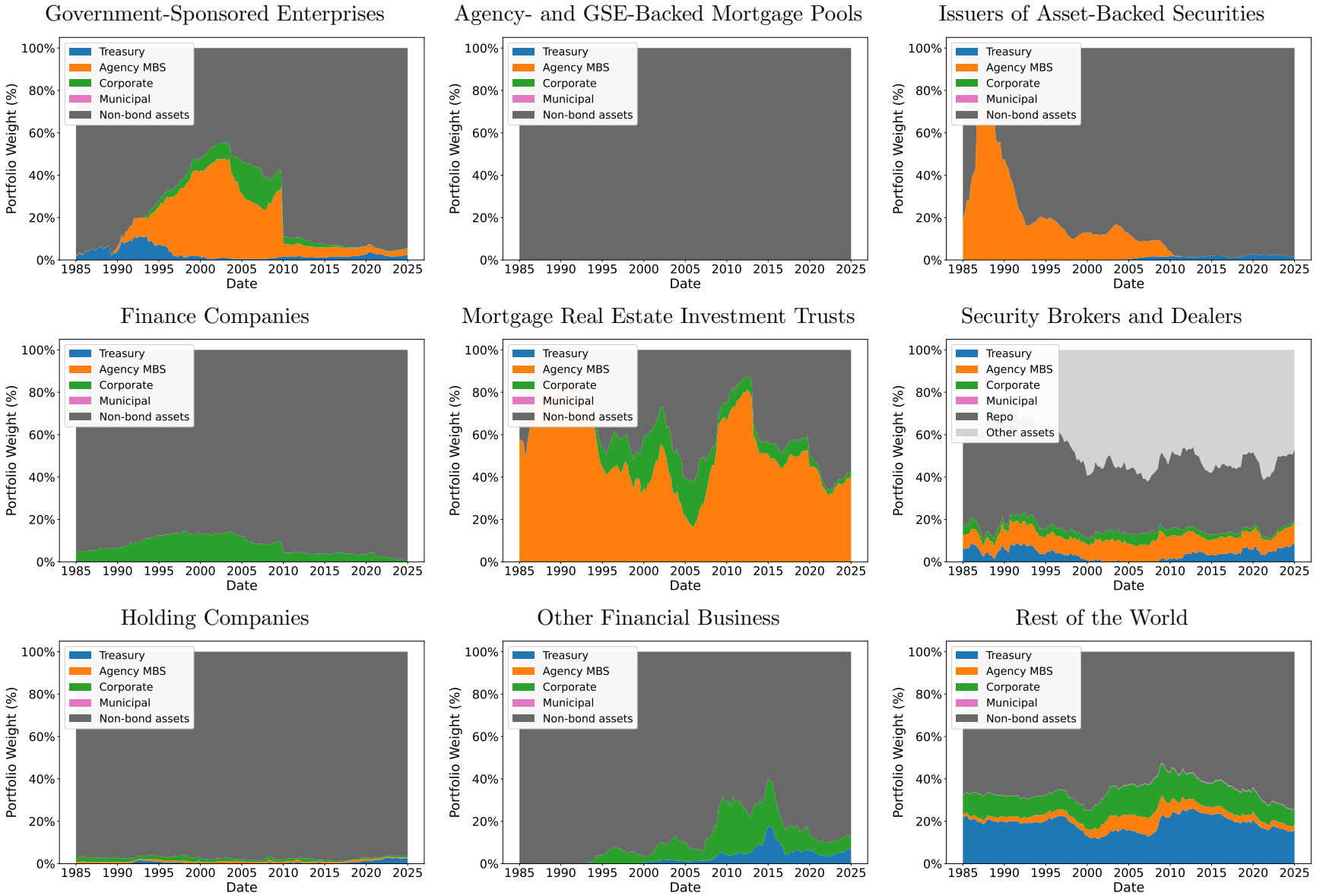
Notes: This figure shows each sector's portfolio as stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the rest of the balance sheet, outside the demand system, and is absent where the outside asset is the total non-bond residual.

Figure A.1: **Portfolio Shares by Sector (continued)**



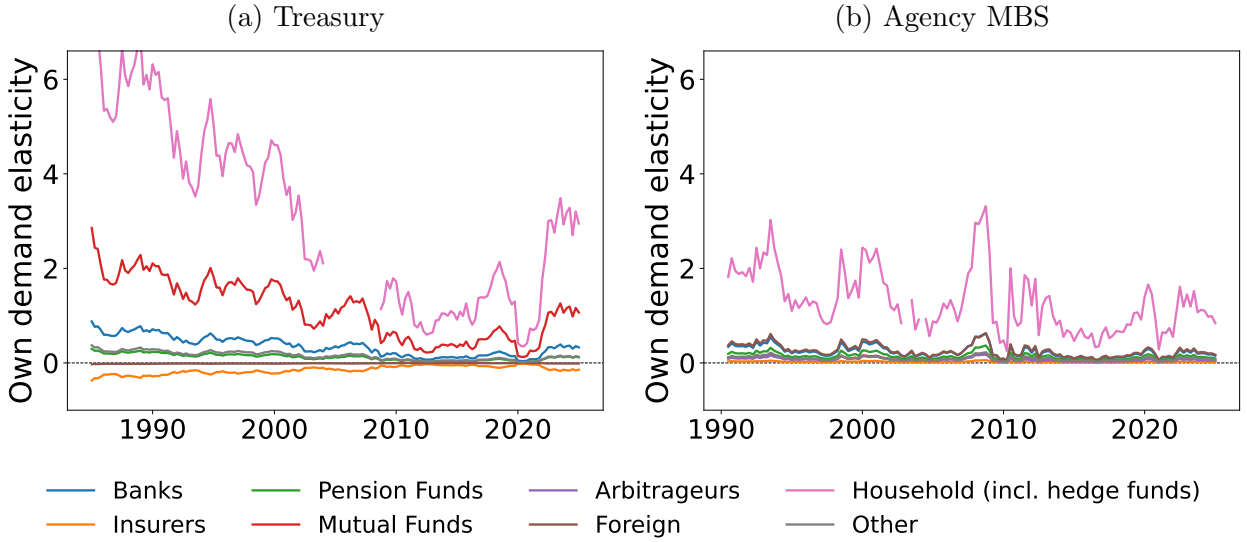
Notes: This figure shows each sector's portfolio as stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the rest of the balance sheet, outside the demand system, and is absent where the outside asset is the total non-bond residual.

Figure A.1: **Portfolio Shares by Sector (continued)**



Notes: This figure shows each sector's portfolio as stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the rest of the balance sheet, outside the demand system, and is absent where the outside asset is the total non-bond residual.

Figure A.2: **Own Demand Elasticity by Sector Type**



Notes: This figure shows the own elasticities $\eta_{it}(n,n) = \beta_i(n)(1 - w_{it}(n))\, y_t(n)$, averaged within sector type with weights $A_{it} w_{it}(n)$, for the logit cells; the two cells of Section 3.7, money market funds and the dealer Treasury book, are excluded, so the arbitrageurs have no Treasury line. Both panels share one scale; the household residual's Treasury line leaves the frame before 1990.

C Tables

Table A.1: **Data Sources for Sector-Level Portfolio Holdings**

Type	Sector	Source
Banks	U.S.-Chartered Depositories	L.111
	Foreign Banking Offices	L.112
	Credit Unions	L.114
	Banks in Affiliated Areas	L.113
Insurers	Life Insurers	L.116
	Property-Casualty Insurers	L.115
Pension Funds	Private Pension Funds	L.118
	State & Local Govt. Retirement Funds	L.120
	Federal Govt. Retirement Funds	L.119
Mutual Funds	Mutual Funds	L.122
	Exchange-Traded Funds	L.124
	Money Market Funds	L.121
	Closed-End Funds	L.123
Arbitrageurs	Brokers and Dealers	L.130
	mREITs	L.129
Foreign	Foreign Official	CSLT
	Foreign Private	CSLT
Household (incl. hedge funds)	Households & Nonprofits	L.101
Other	Nonfinancial Business	L.102
	Government-Sponsored Enterprises	L.125
	Holding Companies	L.131
	State & Local Governments	L.107
	Federal Government	L.106
	Agency/GSE Mortgage Pools	L.126
	Finance Companies	L.128
	ABS Issuers	L.127
	Other Financial Business	L.132

Notes: This table shows the Financial Accounts (Z.1) table codes of each sector's holdings; transactions come from the matching flow (FU) tables. The rest of the world is split into Foreign Official and Foreign Private holders over 1985–2025 with the Federal Reserve's CSLT estimates of cross-border positions (Bertaut and Judson, 2014), which the Financial Accounts do not provide.

Table A.2: **Data Sources for Asset Yields and Characteristics**

Asset	Variable	Source
Treasury	Yield	10-year constant-maturity yield, `DGS10`, FRED
	Time to maturity	`YAS_MTY_YEARS`, Bloomberg (`LUATTRUU`)
	Modified duration	`YAS_RISK`, Bloomberg (`LUATTRUU`)
Agency MBS	Yield	10-year Treasury yield plus Bloomberg US MBS index OAS (`LUMSTRUU`)
	Time to maturity	`WAL_TO_MATURITY`, Bloomberg (`LUMSTRUU`)
	Modified duration	`YAS_RISK`, Bloomberg (`LUMSTRUU`)

Notes: This table shows the data sources. Yields are end-of-quarter; maturity and duration come from the Bloomberg total-return index of each asset class (tickers in parentheses; unhedged, USD); sector transactions and holdings are the Z.1 flow (FU) and level (FL/LM) tables.

Table A.3: **Market Demand Elasticities**

	Sample average		2025Q1	
Price shocked	Treasury	Agency MBS	Treasury	Agency MBS
Treasury (10-year yield)	+0.452	-0.010	+0.533	-0.009
Agency MBS (option-adjusted spread)	-0.016	+0.208	-0.008	+0.186

Notes: This table shows the market elasticity matrix of Figure 3, averaged over the sample and at the 2025Q1 composition; rows index the price that moves, columns the demand that responds. Treasury entries are with respect to the 10-year yield and agency MBS entries with respect to the option-adjusted spread, so the two are not comparable in level (the semi-elasticities are in Table 5).

D Specification Discussion

D.1 Outside-Asset Assignment, Sector by Sector

The demand system measures each sector's bond demand relative to a single outside asset, the non-bond class the sector actually trades off against bonds (Table 1). This subsection records the rationale for each assignment, organized by outside asset.

Loans: U.S.-chartered depositories and credit unions. A depository's marginal portfolio decision is between its securities book and its loan book: deposits fund either, and the securities portfolio serves as the buffer that absorbs deposit inflows when loan demand is weak. The QE literature documents exactly this margin: banks receiving reserve inflows under QE expand lending and adjust securities holdings in tandem (Rodnyansky and Darmouni, 2017; Kandrac and Schlusche, 2021; Kurtzman et al., 2022). Measuring bank bond demand against total non-bond assets would instead lump in reserves and interbank claims, which move with monetary policy mechanically rather than through portfolio choice.

Corporate equities: insurers, pension funds, mutual funds, and households. For the long-term investors the relevant margin is the stock–bond allocation. Insurers and pension funds manage fixed-income portfolios against equity allocations under asset–liability and

return targets; mutual funds, together with ETFs and closed-end funds, reallocate between bond and equity exposures through both manager choice and investor flows; and household financial portfolios trade off directly held bonds against directly held equities. This is the classic rebalancing margin of the portfolio-balance channel: when QE compresses bond yields, these investors are pushed toward equities, and the outside asset is defined to capture precisely that substitution.

Repo: money market funds, broker-dealers, and hedge funds. Money funds allocate between short government paper and repo daily; repo is their actual alternative at the margin, not equities they cannot hold. Dealers likewise manage securities inventory against matched-book and reverse-repo positions, so repo is the financing margin along which their bond holdings expand and contract. Hedge funds enter the data in Treasuries only, where the Enhanced Financial Accounts data are informative and where the cash–futures basis trade concentrates their holdings. That trade is repo-funded: the fund buys the cash Treasury and finances it in the repo market, so repo, not equities or other bonds, is the margin along which its Treasury book expands and contracts, the same financing margin as dealers and money funds. The main 27-sector panel keeps hedge funds inside the household residual, because a full carve-out is infeasible: Form PF includes offshore-domiciled funds, whose holdings the Z.1 assigns to the rest of the world, so subtracting the full hedge-fund book from the residual double-counts the offshore share and produces negative household holdings. The household residual's Treasury slope in Table 4 is therefore the slope of a sector that, after 2010, contains the basis-trade book, and Section 4 reads it that way.

Non-bond U.S. securities: foreign investors. The rest of the world's U.S. portfolio decision is between U.S. bonds and other U.S. securities, chiefly equities; the TIC-based holdings data are organized along exactly this line. Reserve managers, the dominant foreign Treasury holders, allocate between Treasuries, agencies, and other dollar assets, so the within-U.S. margin is the relevant one, not substitution back into home-currency assets.

Total non-bond (residual): the remaining sectors. For sectors without a single dominant alternative, foreign banking offices and banks in U.S.-affiliated areas (balance sheets dominated by interoffice claims), federal government retirement funds (holdings concentrated in special-issue Treasuries), mortgage REITs (levered MBS vehicles whose non-MBS assets are small), and the heterogeneous "Other" group, I use the residual of total financial assets net of every bond class the Financial Accounts report separately. Imposing a specific margin on these balance sheets would assert substitution behavior the data cannot support; the residual is agnostic.

Other bonds as a common outside asset. Corporate and municipal bonds are outside the choice set, so they can serve as one outside asset for every sector. That removes the cost Section 3 names: the slopes become comparable across sectors. It changes what the slope measures, however. The slope becomes substitution within fixed income, between Treasuries or agency MBS and credit, rather than substitution out of bonds, and a corporate yield that moves with the Treasury yield blunts it. I re-estimate the system with each sector's corporate and municipal book as its outside asset, keeping the instruments, controls, and cell models unchanged.

The outside asset changes who absorbs more than how much the market absorbs. The household residual's Treasury slope is 0.64 (standard error 0.28) against 0.73, though its Anderson–Rubin set now reaches −0.02; mutual funds fall from 0.28 to 0.11, and foreign investors rise from zero to 0.16. The Treasury aggregate demand slope rises from 0.130 to 0.160 ($t = 2.6$), so the impact per 1 percent of the float falls from 7.7 to 6.3 basis points; the agency MBS slope falls from 0.512 to 0.421, and its impact rises from 2.0 to 2.4. Historical Treasury effects shrink by a fifth to a third, to 6.3 to 7.9 basis points per 1 percent of the float removed against 7.7 to 12.2 (QE1 −36 against −54, QE4 −151 against −186). A $1 trillion runoff raises Treasury yields by 19 basis points rather than 22 and mortgage yields by 26 rather than 27. The mortgage spread moves more: QE1 compresses it by 74 basis points

against 54. The baseline keeps the non-bond margins, because the portfolio-balance channel of QE is the movement of duration out of the bond market, and a within-bond outside asset cannot see it.

D.2 Transactions versus Market Values

The Financial Accounts decompose the change in every level into transactions, revaluations, and other volume changes. The demand system uses the transactions (Section 2). This subsection documents why the alternative, the market-value share with an index-duration adjustment, does not measure active demand in these data, and what it does to the estimates.

The adjustment adds the revaluation back. For a bond position of duration $D_t(n)$, a yield change $\Delta y_t(n)$ (in percentage points) changes the position's market value by approximately $-D_t(n)\,\Delta y_t(n)/100$ even if the sector trades nothing, so the adjusted change

$$\Delta \tilde{q}_{it}(n) \equiv \Delta \ln \frac{w_{it}(n)}{w_{it}(0)} + D_t(n)\,\frac{\Delta y_t(n)}{100} \tag{24}$$

is meant to isolate active rebalancing. Eren et al. (2026) make this correction with a constant eight-year duration; the observed index duration is the natural alternative. The correction is right for a holder whose recorded level moves one for one with market prices at the index duration, and wrong otherwise. If a sector's recorded level does not revalue, the differenced log share is already the active change, and adding $D\,\Delta y/100$ manufactures a positive response of $D/100$ per percentage point, a slope of about 0.05 at the durations of the two indices, for a sector that traded nothing.

Table A.4 measures how much of the market revaluation each sector's recorded level actually contains: the coefficient of the Financial Accounts' own revaluation series on the index-implied revaluation $-D_t(n)\Delta y_t(n)\,H_{i,t-1}(n)/100$, cell by cell. The pass-through is a continuum. It is at or above one for the market-value holders, broker-dealers and open-end mutual funds; near zero for money market funds; and mostly between 0.1 and 0.5 for

the depositories, insurers, and pension funds whose books mix amortized cost, statutory value, and market value. No sector-specific on–off correction reproduces this, and the index duration is the wrong duration for most of them.

Table A.4: **Pass-Through of Market Revaluation into Recorded Levels**

Sector type	Sector	Treasury	Agency MBS
Arbitrageurs	Security Brokers and Dealers	1.50	0.81
Banks	Credit Unions	0.28	0.20
	Foreign Bank Offices	—	—
	Foreign Banking Offices in the U.S.	—	—
	US Depositories	—	0.26
Foreign Investors	Rest of the World	0.36	0.27
Households	Households/Nonprofits	0.71	0.44
Insurers	Life Insurers	0.42	0.23
	Property-Casualty Insurers	0.32	0.15
Mutual Funds	Money Market Funds	-0.03	—
	Mutual Funds	0.83	—
Other	Federal Government	—	—
	Government-Sponsored Enterprises	-0.01	0.25
	Nonfinancial Business	0.33	—
	Other Financial Business	—	—
Pension Funds	Fed Pension Funds	0.27	0.54
	Private Pension Funds	0.40	0.33

Notes: This table shows, for each sector and asset with a revaluation series in the Financial Accounts, the slope of the recorded quarterly revaluation on the index-implied revaluation of the lagged position, $-D_t(n)\,\Delta y_t(n)\,H_{i,t-1}(n)/100$, 1985Q1–2025Q1: one means the position revalues at the index duration, zero that it does not revalue.

The measurement, not the instrument, drives the results. Re-estimating the common-slope sector-type long-run slopes $\beta^{(4)}$ under both measurements of the dependent variable and both Treasury instruments, market-value shares with the adjustment (24) give banks, insurers, and foreign investors small positive slopes (0.07, 0.12, 0.05 under the granular instrument), which would read as slow, weakly positive demand; with transactions the same sectors are at zero or below under either instrument (-0.10, -0.04, -0.05 under the heteroskedasticity instruments), and the arbitrageurs' large negative market-value slope shrinks to a fraction. Only the household residual is positive under transactions, and its slope rises when the instrument is changed. The inert cluster was the adjustment.

The transaction slope and the market-value slope measure different objects, and they are related by an identity. Equation (4) measures the change in units held. The market-value

share ratio changes by the same amount plus the relative price change, which is common to every holder of the asset and equals $-D_t(n)\Delta y_t(n)/100$ up to carry; regressing the transaction growth less the true revaluation on the yield change therefore gives a slope that differs from the transaction slope by $-D/100$, a uniform shift of about -0.05 across sectors. Under that normalization a holder that trades nothing has a slope of $-D/100$, not zero, and the household residual's Treasury slope of 0.73 is the one contrarian position in the market. Market clearing is written in units, and the Fed's transactions shift units, so the counterfactual uses the transaction slope; the identity is why the choice of normalization does not affect it.

One cell is sensitive to how transaction growth is measured, and to how the level is treated. The Financial Accounts' household Treasury position is a residual, total outstanding less every identified holder, and it is negative from 2004Q2 to 2008Q2, reaching $-$\$318 billion, about a tenth of the float, in 2007Q4. Wherever the paper weights sectors by their holdings, the aggregate demand slope, the market elasticities, and the counterfactual, I floor that residual at zero and let the float renormalize. A negative statistical residual would otherwise put a negative weight on the sector's positive slope and turn the aggregate slope negative in six quarters, which is a measurement artifact and not demand; the reported counterfactuals are unaffected, because every quarter they solve at has a positive residual. Dealers' negative Treasury positions are left as reported: a dealer can genuinely be net short, a residual cannot. Separately, the growth rate $\ln(1+F/H_{-1})$ computed on that near-zero base is extreme in the surrounding quarters (its standard deviation is 0.70 over 2000–2008 against 0.09 before and 0.27 after). Without the 1/99 winsorization of Section 2 the household residual's Treasury slope is 1.21 (standard error 0.36); with it, 0.73 (0.28); and with a 5/95 rule, 0.73 (0.26). The paper uses the 1/99 rule, fixed in advance for every cell, and the Treasury counterfactuals inherit this cell's range.

D.3 Estimation Window

Re-estimated on the 2010Q1–2025Q1 window alone, the long-run slopes of Table 4 split by market. The agency MBS absorbers keep their signs: the household residual at 1.03 (standard error 0.80) against 2.33 on the full sample, banks at 0.33 (0.23) against 0.48, and the arbitrageurs at 0.37 (0.26) against 0.15. The Treasury absorber does not: the household residual's slope is −0.84 with a standard error of 0.82, and no other sector type's Treasury slope is significant in the short window either. The Treasury counterfactuals of Section 5 therefore rest on the pre-2010 behavior of the residual sector, the dependence that Panel (b) of Table 8 quantifies.

D.4 Construction of the Purchase-Path Revisions

The instrument is built to isolate the unexpected part of the Fed's balance-sheet policy. The Fed's realized purchases are not a surprise: once a program is announced it runs on a published monthly schedule, so a quarter's realized purchases are known to the market before the quarter begins. What the market could not anticipate is a change in that schedule, and the instrument is built from the communications that made one.

Events. I read the FOMC statements, minutes, and implementation notes between November 2008 and May 2024 and keep the 21 communications that set or restated the announced path of System Open Market Account purchases or runoff; four restated it and are coded zero. Table A.5 lists them. Communications that restated the path, or changed only the timing of a program without changing the amount purchased over the following year, are kept in the record and coded zero.

Expected and unexpected purchases. For each event e and each asset n, agency MBS or Treasuries, let $P_e^{\text{new}}(n)$ be net purchases over the four quarters following the communication under the newly announced path, and $P_e^{\text{old}}(n)$ the same quantity under the path in

force before it, which is the path announced at the previous event. The revision is

$$R_e(n) = P_e^{\text{new}}(n) - P_e^{\text{old}}(n). \tag{25}$$

The expected path is thus the one the Fed last announced: the market is assumed to expect the Fed to do what it last said it would do, and the whole change is the surprise. 19 of the 21 communications state an explicit total or monthly pace, so $R_e(n)$ follows from the announced amounts alone. The announcement of $40 billion a month of agency MBS in September 2012, for example, is a revision of $480 billion over the following year, and the $600 billion of Treasury purchases in November 2010 is a revision of $600 billion. Two communications name no amount and require inference: the June 2013 press conference, which said that purchases would begin to slow later in 2013 and end by mid-2014, and the March 23, 2020 statement, which committed to purchases "in the amounts needed". For the first I compute the reduction implied by that timetable relative to the pace then in force; for the second I use the purchases actually made through June 2020. Dropping both leaves the first stage essentially unchanged.

Timing and scaling. A change often becomes public before the FOMC statement that formalizes it, through a Jackson Hole speech, a press conference, or the minutes of an earlier meeting. I date each revision to the quarter in which the change first became public, taken from the documented prior communication; five events are dated to an earlier communication, four of them to an earlier quarter, and the "Dated to" column of Table A.5 records them. Revisions dated to the same quarter are summed and scaled by the amount outstanding at the end of the previous quarter,

$$Z_t(n) = \frac{\sum_{e \in t} R_e(n)}{S_{t-1}(n)}, \tag{26}$$

so the instrument measures the surprise as a share of the market. The instrument is set to zero in the 12 crisis quarters of Section 3.6, which removes six communications, the November 2008 announcement, the September 2011 reinvestment decision, both March 2020 announcements, the June 2020 restatement (a zero revision), and the May 2022 runoff caps, each of which arrived during a flight to quality.

What the construction assumes. Treating the previously announced path as the expectation overstates the surprise when the market had priced a change from speeches or data that the record does not capture. Dating each change to its first public signal removes the documented cases. Any anticipation that remains makes part of the measured revision expected, which weakens the first stage rather than creating a spurious one. The exclusion restriction is that a revision affects sector demand only through the yield, conditional on the macroeconomic controls; the checks below show that the revisions do not respond to the previous quarter's yields or macroeconomic conditions.

D.5 Weak-Instrument Checks

Table A.6 reports three checks on the identification. The upper panel re-estimates the Treasury cells with all three heteroskedasticity donors, maturity, issuance, and coupon, in place of the single donor of the baseline. The three-donor block reaches a first stage of 6.4 against the baseline's 7.7, because the added interactions carry no relevance and only dilute the statistic, and the long-run slopes are essentially unchanged: the household residual's Treasury slope is 0.57 ($[0.27, 1.56]$) with three donors against 0.73 ($[0.18, 1.91]$) with the coupon donor alone. The bottom panel shows why the agency MBS block does not use heteroskedasticity instruments: adding them to the Fed's transactions and the Drukker surprise raises the first stage to 17.3, but the overidentification test rejects the joint set for insurers, mutual funds, and the arbitrageurs, because the added relevance is the maturity interaction, and the index maturity of agency MBS moves with prepayment and hence with

Table A.5: **FOMC Communications That Revised the Purchase Path**

FOMC date	Dated to	Agency MBS revision ($bn)	Treasury revision ($bn)	Coding	Communication
Nov 25, 2008	2008Q4[c]	+500	0	explicit	LSAP announced: up to $500bn agency MBS
Mar 18, 2009	2009Q1	+750	+300	explicit	MBS raised to $1.25tn; $300bn Treasuries added
Sep 23, 2009	2009Q3	0	0	explicit	Completion extended to 2010Q1 (timing only)
Aug 10, 2010	2010Q3	0	+253	explicit	MBS principal reinvested in Treasuries
Nov 03, 2010	2010Q3	0	+600	explicit	QE2: $600bn Treasuries
Sep 21, 2011	2011Q3[c]	+253	−253	explicit	MBS principal reinvested in MBS
Sep 13, 2012	2012Q3	+480	0	explicit	QE3: $40bn/month agency MBS
Dec 12, 2012	2012Q4	0	+540	explicit	$45bn/month Treasuries added
Jun 19, 2013	2013Q2	−200	−225	inferred	Taper to start in 2013, end mid-2014
Dec 18, 2013	2013Q4	+133	+175	explicit	Taper begins
Oct 29, 2014	2014Q4	0	0	explicit	Purchases end as scheduled
Jun 14, 2017	2017Q1	−72	−108	explicit	Runoff caps published
Sep 20, 2017	2017Q3	0	0	explicit	Runoff starts October 2017
Mar 20, 2019	2019Q1	0	+340	explicit	Treasury runoff to end September 2019
Mar 15, 2020	2020Q1[c]	+437	+263	explicit	$200bn MBS and $500bn Treasuries
Mar 23, 2020	2020Q1[c]	+560	+1,289	inferred	Purchases "in the amounts needed"
Jun 10, 2020	2020Q2[c]	0	0	explicit	Pace codified, unchanged
Nov 03, 2021	2021Q3	−375	−750	explicit	Taper announced
Dec 15, 2021	2021Q4	−45	−90	explicit	Taper pace doubled
May 04, 2022	2022Q2[c]	−368	−630	explicit	QT caps from June 2022
May 01, 2024	2024Q1	0	+420	explicit	Treasury cap cut from $60bn to $25bn/month

Notes: This table shows, for each FOMC communication, the revision to expected net purchases over the following four quarters relative to the previously announced path, in billions of dollars (negative for lower purchases or faster runoff), and the quarter it is dated to, when the change first became public; [c] marks a crisis quarter, where the instrument is set to zero. "Explicit" events state an amount or pace; "inferred" revisions are computed as described in the text.

the yield. LIML and Fuller estimates coincide with two-stage least squares in every cell.

Table A.6: **Weak-Instrument Checks**

		Single donor (baseline)		Three donors	
Asset	Sector type	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set
Treasury: first-stage F		7.7		6.4	
	Banks	+0.09 (0.11)	[-0.06, 0.36]	+0.03 (0.10)	$\emptyset$
	Insurers	-0.04 (0.06)	[-0.16, 0.14]	-0.05 (0.06)	[-0.20, 0.15]
	Pension Funds	+0.03 (0.04)	[-0.04, 0.11]	+0.03 (0.03)	[-0.04, 0.13]
	Mutual Funds	+0.28 (0.12)	[0.05, 0.65]	+0.28 (0.13)	[-0.04, 0.69]
	Arbitrageurs	-1.67 (1.14)	[-8.52, 0.16]	-1.44 (0.96)	[-9.09, -0.29]^{u}
	Foreign	-0.00 (0.06)	[-0.10, 0.12]	-0.01 (0.07)	[-0.18, 0.14]
	Household (incl. hedge funds)	+0.73 (0.28)	[0.18, 1.91]	+0.57 (0.25)	[0.27, 1.56]
	Other	+0.04 (0.14)	[-0.26, 0.34]	+0.05 (0.14)	[-0.28, 0.30]
Agency MBS: SOMA + Drukker ($F = 3.1$) vs. +Lewbel ($F = 17.3$)		$\beta^{(4)}$	Sargan p	$\beta^{(4)}$	Sargan p
	Insurers	+0.60 (0.33)	0.39	+0.09 (0.08)	0.037
	Foreign	+0.77 (0.48)	0.09	+0.30 (0.18)	0.233
	Mutual Funds	+2.66 (1.44)	0.23	+0.14 (0.28)	0.000
	Arbitrageurs	+1.70 (1.18)	0.64	-0.03 (0.25)	0.018

Notes: This table shows, in the upper panel, long-run Treasury slopes $\beta^{(4)}$ with Newey–West standard errors and 95 percent Anderson–Rubin sets under the paper's instruments and with all three heteroskedasticity donors (maturity, issuance, coupon) in place of the coupon donor ($\emptyset$: empty set; ^{u}: set reaching the grid edge), and in the lower panel agency MBS slopes for the absorber cells under the Fed's transactions and the Drukker (2025) surprise, with and without the agency MBS heteroskedasticity instruments, with overidentification p-values.

Crisis Quarters. The crisis quarters are defined by a criterion rather than by fit: the quarters of the flight-to-quality episodes documented in the literature, listed in Section 3.6. The August 2011 quarter is the one that matters for the estimates. It contains the U.S. downgrade and the escalation of the euro-area crisis, and it also contains the September 2011 decision to reinvest agency MBS principal in agency MBS, a large purchase revision in a quarter when the mortgage spread widened by 43 basis points. Its quarterly VIX, 30.6, exceeds that of 2008Q3 and of every 2022 quarter already in the set, and a mechanical VIX rule includes it under every threshold I tried. A mechanical VIX rule is not a good substitute for the criterion, however: it also adds the 1998–2003 and 2009 quarters and drops part of 2022, and under it the agency MBS instrument moves the Treasury yield as well, failing the cross-asset placebo, while the Treasury block loses more than half of its first stage.

The Menu of Candidate Instruments for Agency MBS. Table A.7 reports every candidate I examined for the agency MBS block, each run three ways: on the option-adjusted spread, which is the price the block estimates; on the Treasury yield, which is the cross-asset placebo an asset-specific supply shifter must fail to move; and on the MBS yield level, the price a yield specification would use. Read the first two column pairs together. The Fed's realized transactions paired with the Drukker surprise move the spread with an F of 3.1 but move the Treasury yield more (4.6), the wrong ordering for an MBS-specific instrument; the transactions alone have the right ordering but a first stage of only 4.2. The purchase-path revision over all quarters moves the spread at 5.3 and the Treasury yield at 12.0 ($t = -3.46$): in the crisis quarters the announcement and the flight to quality are the same event. Zeroed in those quarters the ordering is right and the gap is wide, 18.3 on the spread against 1.0 on the Treasury yield (18.1 in Table 3, which uses the second stage's control set and kernel). Dropping the two judgment-coded events or the 2020–2025 quarters leaves the ordering intact.

The other candidates fail for identifiable reasons. Gross agency issuance at par is strongly correlated with the MBS yield but with a negative sign: issuance rises when rates fall because borrowers refinance, so the series traces the supply curve rather than shifting it, and it belongs in the supply estimation of Section 5.2. Foreign official purchases of agency securities and the retained portfolios of the government-sponsored enterprises move with the MBS yield level (F of 6.6 and 10.2) but not with the spread, and the foreign series has the positive sign of buying after yields rise. The composition instrument, the MBS revision less the Treasury revision, has little power for the spread (2.4), and the LSAP factor of Swanson (2021) does not survive quarterly aggregation (0.3). The Treasury coupon instrument has no power inside the MBS block in the baseline control set, because the contemporaneous MBS index maturity moves with the rate level through prepayment and absorbs the level variation; entering that control with a one-quarter lag restores its relevance ($F = 19.3$), which is how the level-versus-spread restriction of Section 4 is tested.

Table A.7: **Candidate Instruments for the Agency MBS Block**

	MBS spread		Treasury yield (placebo)		MBS yield	
Candidate instrument	F	t	F	t	F	t
SOMA transactions + Drukker (2025) surprise	3.1	-2.05	4.6	+1.51	3.0	-2.02
SOMA transactions alone	4.2	-2.05	2.3	+1.51	4.1	-2.03
Drukker (2025) surprise alone	1.9	-1.37	6.7	-2.59	3.1	-1.77
Swanson (2021) LSAP factor	0.3	-0.51	0.6	+0.76	0.7	-0.83
Fed purchase-path revision, all quarters	5.3	-2.31	12.0	-3.46	9.0	-3.00
Fed purchase-path revision, crisis-robust	18.3	-4.27	1.0	-0.99	4.0	-1.99
composition revision, MBS less Treasury	2.4	-1.56	7.4	-2.71	0.4	-0.60
path revision, Treasury leg partialled out	5.4	-2.32	12.4	-3.52	2.8	-1.69
path revision + SOMA + Drukker	6.5	-2.39	8.3	-3.96	5.9	-2.96
Foreign official agency transactions (CSLT)	1.2	+1.08	0.4	+0.63	6.6	+2.56
GSE retained-portfolio transactions	0.9	-0.93	0.4	+0.64	10.2	-3.19
Treasury coupon Lewbel applied to agency MBS	2.3	+1.51	14.0	-3.74	3.4	+1.84

Notes: This table shows first-stage F and t statistics (Newey–West, four lags) of each candidate instrument on the quarterly change in the agency MBS option-adjusted spread, the 10-year Treasury yield (the cross-asset placebo), and the agency MBS yield, 1990Q3–2025Q1, with the second-stage controls. The bold row is the paper's instrument.

Dating the Revisions: the Committee Signals Before It Votes. Dated to the FOMC statement that formalizes it, the Treasury leg of the purchase-path revisions has no first stage, with a positive sign ($t = +0.60$), meaning yields rise when the Fed announces that it will buy more. The event quarters look perverse. The Fed announced $600 billion of Treasury purchases in 2010Q4 and the 10-year yield rose 77 basis points over that quarter; it announced $300 billion in 2009Q1 and the yield rose 46; it added $45 billion a month in 2012Q4 and the yield rose 13.

The reason is anticipation. Regressing the quarterly change in the 10-year term premium on the statement-dated revision at t, $t+1$, and $t-1$ together, the contemporaneous coefficient is zero ($t = 0.18$) while the coefficient on the *next* quarter's revision is negative ($t = -1.73$): the market prices part of a Treasury program before the Committee votes on it. The agency MBS leg is different, with a large contemporaneous coefficient ($t = -4.51$) and no lead ($t = -0.64$): what gets telegraphed is the size of a program, and the composition is the part that surprises.

I therefore date each revision to the quarter in which the change first became public,

taken from the documented prior communication; Table A.5 lists the five events dated to an earlier communication. Under this dating the Treasury revision loads on the 10-year term premium with $t = -2.57$ and on the yield with $t = -1.55$, against $+0.25$ and $+0.60$ under statement dating.

Two checks support using it. First, the revisions are not a reaction to the previous quarter: regressed on the prior quarter's yield change, term-premium change, output growth, inflation, and volatility, the Treasury revision has an R^2 of 0.11, and instrumenting with only its unpredictable part leaves the first stage of the pair unchanged. Second, the Treasury revision moves the mortgage spread only with the opposite sign ($t = +2.17$), so it cannot be what identifies the agency MBS block. Paired with the coupon instrument the Treasury block has a joint F of 7.7, no sector type rejects the overidentifying restriction (minimum p of 0.30), and the Anderson–Rubin sets narrow.

Which Instrument Identifies the Treasury Block? The Treasury block uses two instruments, and a natural question is whether the internal one is needed at all. Table A.8 reports the block under each instrument alone, under the two external instruments together, and under the paper's pair. A third external candidate appears there as well: Japanese foreign-exchange intervention. When the Ministry of Finance sells yen and buys dollars it invests the proceeds in Treasuries, so an intervention shifts foreign demand for a reason that originates in Japanese exchange-rate policy. It is the strongest single instrument in the table, with a first stage of 16.4 and the sign a demand shift implies, and it moves the mortgage spread only weakly and with the opposite sign.

The instruments identify off different parts of the sample and different sectors. The coupon interaction has a first stage of 12.9 before 2009 and 0.2 after: the calm-period heteroskedasticity it exploits is a pre-crisis phenomenon. The external instruments are the mirror image, at 7.9 for the purchase-path revision and 6.7 for the Japanese intervention after 2009. They also disagree about who absorbs. The coupon instrument delivers the household

residual at 0.76 and mutual funds at 0.28, the long-duration buy-and-hold absorbers; the external instruments deliver banks, at 0.51 with a set of $[0.05, 1.15]$, and leave the household residual at 0.77 with an unbounded set. That is a difference between periods rather than a conflict: the overidentification test rejects nowhere in the paper's pair.

The aggregate demand slope settles the question. At the 2025Q1 composition it is 0.130 (standard error 0.049) under the paper's pair and 0.132 (0.051) under the coupon instrument alone, both distinguishable from zero. Under the two external instruments together it is 0.156 with a standard error of 0.159, and under either of them alone it is not distinguishable from zero: the purchase-path revision alone gives 0.105 with a standard error of 0.320 and the Japanese intervention alone 0.191 with a standard error of 0.168. The household residual and the mutual funds carry most of the aggregate demand slope between them, and only the internal instrument identifies those two cells. What the comparison establishes is narrower and more useful than a ranking: the *level* of the Treasury aggregate demand slope is similar however it is identified, and the *identity* of the marginal buyer is not.

Table A.8: **Treasury Demand by Instrument**

	Coupon Lewbel		Fed path revision		Japan FX intervention		External only		Baseline	
Sector type	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set	$\beta^{(4)}$	AR set
Banks	+0.07	[-0.13, 0.30]	+0.77	[-0.21, 2.30]	+0.48	[0.05, 1.01]	+0.51	[0.05, 1.15]	+0.09	[-0.06, 0.36]
Insurers	-0.04	[-0.14, 0.09]	+0.04	[-0.81, 0.67]	-0.00	[-0.27, 0.33]	+0.00	[-0.31, 0.37]	-0.04	[-0.16, 0.14]
Pension Funds	+0.03	[-0.02, 0.11]	-0.17	[-1.03, 0.44]	+0.08	[-0.07, 0.25]	+0.06	[-0.12, 0.26]	+0.03	[-0.04, 0.11]
Mutual Funds	+0.28	[0.05, 0.55]	+0.40	[-0.41, 1.37]	-0.09	[-0.79, 0.55]	-0.04	[-0.43, 0.67]	+0.28	[0.05, 0.65]
Arbitrageurs	-1.72	[-11.95, -0.18]^u	-1.31	[-6.49, 1.39]	-18.73	[-1245.32, 1207.87]^u	-1.41	[-10.66, 7.84]^u	-1.67	[-8.52, 0.16]
Foreign	-0.01	[-0.14, 0.11]	+0.33	[-0.09, 2.06]	-0.02	[-0.20, 0.18]	+0.01	[-0.14, 0.21]	-0.00	[-0.10, 0.12]
Household (incl. hedge funds)	+0.76	[0.29, 1.59]	-0.17	[-5.47, 5.95]	+1.06	[-7.55, 9.66]^u	+0.77	[-4.36, 8.96]^u	+0.73	[0.18, 1.91]
Other	+0.05	[-0.20, 0.34]	-0.48	[-1.96, 0.86]	+0.26	[-0.40, 0.99]	+0.19	[-0.61, 0.79]	+0.04	[-0.26, 0.34]
First-stage F (ten-year yield)	14.0		3.5		16.4		8.8		7.7	
Aggregate demand slope, 2025Q1	+0.132 (0.051), t 2.6		+0.105 (0.320), t 0.3		+0.191 (0.168), t 1.1		+0.156 (0.159), t 1.0		+0.130 (0.049), t 2.7	

Notes: This table shows long-run Treasury slopes $\beta^{(4)}$ by sector type from Equation (10), 1985Q1–2025Q1, under five instrument sets: the coupon heteroskedasticity instrument; the Fed Treasury purchase-path revision; Japanese foreign-exchange intervention; the two external instruments together; and the coupon instrument with the path revision, the paper's set. 95 percent Anderson–Rubin sets appear beside each estimate (^{u}: set reaching the grid edge). The last two rows are the first stage on the 10-year yield and the implied 2025Q1 aggregate demand slope with its delta-method standard error, under the cell models of Section 3.7.

D.6 What the Household Residual Holds

The Treasury counterfactual rests on one cell, and the paper asserts that after 2010 that cell is mostly the hedge funds running the cash–futures basis trade. The Federal Reserve's

Enhanced Financial Accounts report hedge-fund balance sheets from Form PF quarterly since 2013 and let the assertion be checked.

Over 2013Q1–2025Q1 hedge funds hold 122 percent of the residual's coupon Treasuries on average, 99 percent at the median and 85 percent in the last quarter, and the two levels correlate at 0.93. In flows, the residual's Treasury transactions regress on the quarterly change in hedge-fund holdings with a coefficient of 0.62 (Newey–West standard error 0.17, $t = 3.7$). The residual and the basis trade are the same position.

They cannot, however, be separated into two sectors and estimated jointly, which would be the natural way to add observations to the cell that carries the aggregate. The residual net of hedge funds is negative in 24 of the 49 quarters. The two series are not on the same basis: the Financial Accounts figure is a *net* residual, total outstanding less every identified holder, while the Form PF figure is a *gross* long financed by repo, and a repo-financed long is also in the Accounts as the lender's holding, so differencing one from the other double-counts the financing leg.

The basis trade can be netted out with a proxy, if not with the Form PF levels. Its cash leg is hedged with a short in Treasury futures, and the Commodity Futures Trading Commission's Traders in Financial Futures report gives the leveraged funds' net short in Treasury futures weekly since 2010Q3; the notional of that short tracks the hedge-fund Treasury holdings of the Enhanced Financial Accounts (correlation 0.95 over 2013Q1–2025Q1) and the residual's level (0.97), and it reaches $948 billion in 2025Q1, a third of the residual. Table A.9 re-estimates the cell with the residual's level net of that proxy and its transactions net of the proxy's change. The slope does not move: 0.76 against 0.73 on the full sample, because the slope is identified before 2010, when the proxy is zero, and because the basis trade's own four-quarter response to the instrumented yield change is not distinguishable from zero (1.33, standard error 1.06, on 2010Q4–2025Q1). The 2010–2025 window narrows but stays uninformative, -0.56 with a standard error of 0.55 and an Anderson–Rubin set that contains the full-sample value.

Table A.9: **The Household Residual Net of the Basis Trade**

	Residual as recorded		Residual net of the basis trade	
Window	$\beta^{(4)}$ (s.e.)	AR set	$\beta^{(4)}$ (s.e.)	AR set
1985Q1–2025Q1	+0.73 (0.28)	[0.18, 1.91]	+0.76 (0.26)	[0.23, 1.92]
2010Q1–2025Q1	-0.84 (0.82)	[-2.15, 0.80]	-0.56 (0.55)	[-1.22, 1.21]

Notes: This table shows the household residual's long-run Treasury slope $\beta^{(4)}$ from Equation (10) with the paper's instruments and controls, by window, with Newey–West standard errors and 95 percent Anderson–Rubin sets. "Net of the basis trade" subtracts the leveraged funds' net short in Treasury futures (CFTC Traders in Financial Futures, 2-year through Ultra Bond contracts at notional, last report of each quarter, from 2010Q3) from the residual's Treasury level, and its quarterly change from the transactions, before rebuilding the transaction growth of Equation (4); every other cell and the counterfactual's float weights are unchanged.

What follows is a statement about interpretation rather than a correction. The slope the counterfactual applies to a hedge-fund position is identified where the same residual was households and nonprofits: the 2010–2025 window on its own gives -0.84 with a standard error of 0.82, an interval that contains every value the counterfactual could use. A levered arbitrageur and a buy-and-hold household are not obviously the same demand curve, and the lower Anderson–Rubin row of Panel (b) of Table 8, 45 basis points against 22, is the case in which the fund absorbs less readily than the household did.

D.7 The Household Residual over the Business Cycle

Every QE program ran in or just after a recession, and a demand system with one slope per cell prices them with a slope estimated mostly on expansions. This appendix adds one parameter, and only to the cell that needs it.

I screened all 16 cells against three definitions of bad times, the NBER recession dates, quarters with the VIX in its top quintile, and quarters with negative real GDP growth, estimating

$$y_{i,t}^{(4)} \;=\; \beta_i^{G}\,\Delta y_t \;+\; \delta_i\left(\Delta y_t \times \mathrm{REC}_t\right) \;+\; \text{controls} \;+\; u_{it},$$

with each block's instruments entered twice, as Z_t and $Z_t \times \mathrm{REC}_t$, so both the level and

the interaction are instrumented. Under the NBER dates two Treasury cells move. The household residual's slope is $\beta^G = 0.34$ in expansions against $\beta^G + \delta = 3.58$ in recessions, with $\delta = 3.24$ and a standard error of 1.75, so $p = 0.064$; foreign investors move the other way, from $+0.06$ to -0.16 ($p = 0.04$), as reserve managers step back when the household residual steps in. The household residual's agency MBS slope does not move, $+0.69$ on a standard error of 1.30, so the MBS block is left alone.

The household cell matters most because it carries 82 percent of the Treasury aggregate demand slope at the 2025Q1 composition. Re-running the historical episodes with each program's household slope set to the slope of its own window, weighted by the share of quarters the NBER dates as a recession, changes the recession programs most. The two recession programs fall into line with the literature: QE1 goes from -54 to -26 basis points against -25 in Eren et al. (2026), and QE4 from -186 to -146 against -121. QT1, which ran entirely in an expansion, goes from $+23$ to $+34$ against $+43$.

The restriction is not free, and the paper does not adopt it as the baseline. The expansion slope it implies, 0.34, is half the pooled 0.73 and carries a t of 1.18, so the programs that ran in expansions become larger and less certain: QE3 moves from -112 to -136 against -57 in the same comparison. Splitting 16 recession quarters out of the 159 in the estimation sample buys a sharp answer for two episodes at the cost of precision everywhere else. What the exercise establishes is narrower and more useful: the asymmetry between QE and QT that Eren et al. (2026) document is in this system too, it lives in one cell, and it is the reason a symmetric specification overstates the recession-era programs.

D.8 Nested Tests of the Slope Structure

The system estimates one slope per sector type and asset. Table A.10 asks whether the data need all 16, by minimum distance against two restrictions: one slope per type, which is the standard demand system, and a type factor times an asset loading. The first is rejected at the 10 percent level ($p = 0.06$) and the second is not rejected ($p = 0.29$), so the asset

dimension matters but the data are content with a single loading ratio, 0.32 on the Treasury yield against 1.38 on the agency MBS spread. The counterfactual uses the unrestricted slopes.

Table A.10: **Nested Tests of the Slope Structure**

Restriction	Parameters	Distance χ^2 (df)	p
Common slope across assets, β_i	8	15.1 (8)	0.057
Proportional, $\beta_i\,\theta(n)$	9	8.5 (7)	0.294
Unrestricted, $\beta_i(n)$	16	0.0 (0)	—

Notes: This table shows minimum-distance tests on the 16 long-run slopes of Table 4, treating the cells as independent with their Newey–West standard errors. "Common slope" equates each type's Treasury and agency MBS slopes (8 parameters); "proportional" sets $\beta_i(n) = \beta_i\,\theta(n)$ (9 parameters, loadings normalized to unit mean square). The statistic is χ^2 with degrees of freedom equal to the number of restrictions.

D.9 Variance Decomposition of the Two Prices

The counterfactuals of Section 5 answer a stock question: what the yield would have been had the Fed's portfolio stayed at its pre-episode level. A complementary flow question, asked of the equity market by Koijen and Yogo (2019), is how much of the observed variation in each price is accounted for by supply, by the growth of the holders' other assets, and by shifts in demand. I answer it as they do, with a sequence of counterfactual equilibria. Over a four-quarter window ending in quarter t, every private holder's cumulative transaction growth in market n is $G_i(n) = G_i(0) + \beta_i(n)\,\Delta^{(4)}y(n) + \Delta d_i(n)$, where $G_i(0)$ is the growth of its outside asset and $\Delta d_i(n)$ collects the characteristics, the macro controls and latent demand, defined as the residual so that the identity is exact. With $L_i(n)$ the holder's position at the start of the window, the market clears when the end-of-window positions sum to the private float,

$$\sum_i L_i(n)\,\exp\big(G_i(0) + \beta_i(n)\,\Delta^{(4)}y(n) + \Delta d_i(n)\big) \;=\; S_t(n). \tag{27}$$

Starting from the start-of-window equilibrium, in which no input has moved and the price change is zero, I switch the inputs on one at a time, the Fed's transactions, then net issuance,

then outside-asset growth, then the demand shifters, and solve Equation (27) for the price change after each step; the last step returns the observed change exactly. The difference between successive solutions is the price change attributable to that input, and the variance share of each input is the covariance of its price change with the observed change over the variance of the observed change, as in Koijen and Yogo (2019); the shares sum to one. Each step is solved by the continuation of Section 5. Where the path folds, which happens for the Treasury yield in 50 of 157 windows, the QE and QT years in which the aggregate slope was thin and the supply step alone has no equilibrium connected to the starting one, the first-order step $1/\sum_i s_i\beta_i(n)$ is used instead and the window is counted in the table. Slopes are the long-run $\beta^{(4)}$ with the cell models of Section 3.7, and each market is solved on its own price.

Table A.11: **Variance Decomposition of Four-Quarter Price Changes**

Variance share (%)	Treasury (10-year yield)	Agency MBS (spread)
Supply	13	17
of which net issuance	30	5
of which Federal Reserve transactions (QE/QT)	-16	12
Outside-asset growth	7	-8
Demand shifters	79	90
Quarters at a fold (first-order step used)	50	0
Aggregate demand slope (mean)	0.09	0.40
Std. dev. of the 4-quarter change (pp)	1.05	0.31
Quarters	157	135

Notes: This table shows the share, in percent, of the variance of each price's four-quarter change accounted for by each input of Equation (27), computed from the sequence of counterfactual equilibria described in the text as the covariance of the input's price change with the observed change over the variance of the observed change, with holdings at the start of each window (Treasury 1985Q1–2025Q1, agency MBS 1990Q3–2025Q1); supply, outside-asset growth and demand shifters sum to 100, and a negative share means the input moved the price against the others. "Federal Reserve transactions" are the Fed's net purchases (negative for runoff) and "net issuance" the remaining change in the private float; "outside-asset growth" is the holders' transaction growth in their outside assets; "demand shifters" is the residual. "Quarters at a fold" counts the windows in which a step has no connected equilibrium and the first-order step is used.

Three results stand out. First, demand shifts account for most of the variation in both prices, 79 and 90 percent, as they do for equity prices in Koijen and Yogo (2019); the

transaction form attributes to the shifters the part of the response still under way at the end of the window as well. Second, supply matters more for the agency MBS spread than for the Treasury yield. Net issuance and the Fed's transactions together account for 17 percent of the variance of the spread, the Fed alone for 12 percent, and no window folds, because the MBS slope is large. For the Treasury yield supply accounts for 13 percent, the sum of a 30 percent share for net issuance and a -16 percent share for the Fed: in the QE decade the thin Treasury slope turns the Fed's purchases and the Treasury's issuance into large offsetting terms, and those are the windows in which the supply step folds. Third, the decomposition understates the Fed. It dates supply by realized transactions, while the stock effect of a program is priced when the program is announced, often quarters before the purchases settle, and that anticipation lands in the demand-shifter term. The counterfactuals of Section 5, which hold the balance-sheet change in place and price it through the estimated slopes, answer the question this accounting cannot: not how much of the realized variance the Fed's flows explain, but how much the yields would have differed without them.